Opinion Perception Survey Analysis (preregistration)

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Invalid Date

Table of contents

THIS VERSION CALCULATES THE WEIGHTS PURELY BASED ON

```
load_or_install <- function(pkg) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  }
}
pkgs <- c(
  "tidyverse", "lme4", "lmerTest", "nloptr", "httpgd",
  "languageserver", "lmtest", "sandwich", "caret", "knitr",
  "prettycode", "performance", "estimatr", "patchwork", "broom"
)
```

```r
library("broom.mixed")
knitr::opts_chunk$set(comment = NA)
invisible(lapply(pkgs, load_or_install))
options(vsc.viewer = TRUE)
theme(
  text = element_text(family = "Noto Sans")
)
options(tibble.print_max = Inf)
options(tibble.width = Inf)
```

1 Hypotheses and analysis plan

All hypotheses concerning a single slope parameter (e.g., H1 to H3) are tested using a t-test at significance level $\alpha = 0.05$. Hypotheses involving multiple parameters are tested using a single F-test on their joint significance, also at $\alpha = 0.05$. All subsequent hypotheses follow this convention without further repetition.

NOTE: I am using all dots to infer the issue weights here. For using just the parties, change the line below to "'fitPartyDots"'

```r
fitmode = 'fitAllDots'
```

```r
# -- Standardised test-result printers ----------------------------------------

#' One-sided t-test result block (for single beta)
print_one_sided <- function(label, beta, se, t, df = Inf,
                            direction = c("negative", "positive"),
                            ci = NULL) {
  direction <- match.arg(direction)
  p_two  <- 2 * pt(-abs(t), df = df)
  p_one  <- if ((direction == "negative" && beta < 0) ||
               (direction == "positive" && beta > 0)) p_two / 2 else 1 - p_two / 2
  supported <- p_one < 0.05 && ((direction == "negative" && beta < 0) ||
                                  (direction == "positive" && beta > 0))

  cat(sprintf("=== %s ===\n", label))
  cat(sprintf("  Estimate  = %8.4f\n", beta))
  cat(sprintf("  SE        = %8.4f\n", se))
  cat(sprintf("  t         = %8.4f\n", t))
  cat(sprintf("  df        = %s\n",    ifelse(is.finite(df), sprintf("%.1f", df), "infty")))
  if (!is.null(ci))
    cat(sprintf("  95%% CI    = [%.4f, %.4f]\n", ci[1], ci[2]))
  cat(sprintf("  p (one-sided, beta %s 0) = %.4f\n",
            ifelse(direction == "negative", "<", ">"), p_one))
  cat(sprintf("  Supported : %s\n\n", ifelse(supported, "YES", "NO")))
  invisible(list(beta = beta, t = t, p_one = p_one, supported = supported))
}

#' LRT / F-test result block
print_lrt <- function(label, stat, df_num, df_den = NULL, p,
                      type = c("chi2", "F")) {
  type <- match.arg(type)
  cat(sprintf("=== %s ===\n", label))
  if (type == "chi2")
    cat(sprintf("  chi^2(%d) = %.3f,  p = %.4f  %s\n\n",
                df_num, stat, p, sig_stars(p)))
  else
    cat(sprintf("  F(%d, %d) = %.3f,  p = %.4f  %s\n\n",
                df_num, df_den, stat, p, sig_stars(p)))
  invisible(list(stat = stat, p = p))
}

#' Model-fit comparison table
print_fit_table <- function(...) {
  models <- list(...)
  tbl <- data.frame(
```

```r
    Model    = names(models),
    AIC      = round(sapply(models, AIC), 1),
    BIC      = round(sapply(models, BIC), 1),
    logLik   = round(sapply(models, function(m) as.numeric(logLik(m))), 1),
    npar     = sapply(models, function(m) length(fixef(m)))
  )
  tbl$delta_AIC <- tbl$AIC - min(tbl$AIC)
  tbl$delta_BIC <- tbl$BIC - min(tbl$BIC)
  print(tbl, row.names = FALSE)
  invisible(tbl)
}

tidy_print <- function(m, mixed = FALSE) {
  tidy_fn <- if (mixed) broom.mixed::tidy else broom::tidy
  tbl <- tidy_fn(m, effects = "fixed") |>
    select(term, estimate, std.error, p.value) |>
    mutate(across(where(is.numeric), \(x) round(x, 3)))
  print(tbl)
  knitr::kable(tbl)
}

sig_stars <- function(p) {
  case_when(p < .001 ~ "***", p < .01 ~ "**", p < .05 ~ "*",
            p < .1 ~ ".", TRUE ~ "ns")
}

# Extract lmer beta, SE, t, Satterthwaite df in one call
lmer_coef <- function(model, term) {
  cf <- summary(model)$coefficients
  list(
    beta = cf[term, "Estimate"],
    se   = cf[term, "Std. Error"],
    t    = cf[term, "t value"],
    df   = cf[term, "df"],          # added by lmerTest
    p2   = cf[term, "Pr(>|t|)"]
  )
}
```

```r
questions_sc <- c(
  "climate_concern", "gay_marriage", "rights_indep_integration",
  "econ_inequality", "regulate_internet", "east_germans"
)
parties         <- c("Left Party","BSW","Green Party","SPD","FDP","CDU/CSU",
                     "AfD","No party","Other party","Refuse to say/No answer")
partiesVars     <- c("LeftParty","BSW","GreenParty","SPD","FDP","CDU/CSU","AfD")
partiesCloseSel <- c("Left Party","BSW","Green Party","SPD","FDP","CDU/CSU","AfD")

# Shared column-name vectors (used across many hypotheses)
col_deltas    <- paste0("deltaX_",       questions_sc)
col_weights   <- paste0("w_",            questions_sc)
col_var_std   <- paste0("std_socialCircle_ops_", questions_sc)
col_var_var   <- paste0("var_socialCircle_ops_", questions_sc)
```

2 Load Data

```r
df_p <- read.csv(
  "processed_data/2026-07-07_data_processed_participant_withAllIssueWeights.csv"
)

inds_bothwaves <- df_p |>
  count(id) |>
  filter(n == 2) |>
```

```r
  pull(id)

treatment_ids <- df_p |>
  filter(wave == 2, treatment_wave2 == TRUE, id %in% inds_bothwaves) |>
  pull(id)
control_ids <- df_p |>
  filter(wave == 2, treatment_wave2 == FALSE, id %in% inds_bothwaves) |>
  pull(id)

df_p <- df_p |>
  mutate(
    overall_std_socialCircle = rowMeans(across(all_of(col_var_std))),
    across(all_of(col_var_std), function(x) x^2,
           .names = "var_{gsub('std_', '', col)}"),
    wave = factor(wave)
  ) |>
    rename_with(~ str_replace(., "alpha_deltaX_", "alpha_"))|>
    rename_with(~ str_replace(., paste0("alpha_", fitmode,"_deltaX_"), paste0("alpha_", fitmode, "_")))


df_diff <- read.csv(
  "processed_data/2026-07-07_data_processed_differences_withAllIssueWeights.csv"
) |>
  mutate(
    ingroupdummy = as.integer(ingroupdummy == "True"),
    wave         = factor(wave)
  ) |>
  left_join(
    df_p |> select(id, wave,
                   all_of(paste0("var_socialCircle_ops_",  questions_sc)),
                   all_of(paste0("std_socialCircle_ops_", questions_sc))),
    by = c("id", "wave")
  ) |>
      rename_with(~ str_replace(., "alpha_deltaX_", "alpha_"))|>
      rename_with(~ str_replace(., paste0("alpha_", fitmode,"_deltaX_"), paste0("alpha_", fitmode, "_")))
```

3 Hypotheses

3.1 Aim 1

Aim 1: Compare approaches to measuring perceived political differences

3.1.1 H1

3.1.1.1 H1. Participants place the typical voter of the party they affiliate with nearer to themselves than other typical voters.

We model the distance between the dots representing the participant (self) and the typical voter of each party r as

$$d_{n,\mathrm{self},r} = \beta_0 + \beta_1 \mathbb{I}(r = party_n) + u_n + \epsilon_{n,r}, \tag{1}$$

where $\mathbb{I}(\cdot)$ is an indicator function, u_n is a participant level random intercept, and $\epsilon_{n,r}$ is an error term.

The prediction is $\beta_1 < 0$. Participants who prefer a political party place the typical voter of that party closer to themselves than the average of the typical voters of other parties.

```r
df_h1 <- df_diff |>
  filter(wave %in% c(1, 2), dot1 == "self", dot2 %in% partiesVars) |>
  select(id, wave, ingroupdummy, party, pixel_dist)

m_h1    <- lmer(pixel_dist ~ ingroupdummy + wave + (1 | id), data = df_h1, REML = TRUE)
ci_h1   <- confint(m_h1, parm = "ingroupdummy", method = "Wald")
cf_h1   <- lmer_coef(m_h1, "ingroupdummy")

print_one_sided("H1: Ingroup party placed closer to self",
```

```
                 cf_h1$beta, cf_h1$se, cf_h1$t, cf_h1$df,
                 direction = "negative", ci = ci_h1)
```

```
=== H1: Ingroup party placed closer to self ===
  Estimate  =  -0.2520
  SE        =   0.0041
  t         = -61.5081
  df        = 13785.6
  95% CI    = [-0.2600, -0.2440]
  p (one-sided, beta < 0) = 0.0000
  Supported : YES
```

3.1.2 H2

3.1.2.1 H2. Distances to voters on the map correspond to likeability ratings.

We model likeability ratings of the typical voter of political party r (rated only in wave 2) as a function of map distances:

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \tag{2}$$

The prediction is $\beta_1 < 0$. Voters placed farther away from the participant's own dot on the map receive lower likeability ratings than voters placed closer to the participant's own dot.

```
df_h2 <- df_diff |>
  filter(wave == 2, dot1 == "self", dot2 %in% partiesVars) |>
  select(id, sympathy, party, pixel_dist)

m_h2  <- lmer(sympathy ~ pixel_dist + (1 | id), data = df_h2, REML = TRUE)
ci_h2 <- confint(m_h2, parm = "pixel_dist", method = "Wald")
cf_h2 <- lmer_coef(m_h2, "pixel_dist")

print_one_sided("H2: Map distance predicts likeability",
                cf_h2$beta, cf_h2$se, cf_h2$t, cf_h2$df,
                direction = "negative", ci = ci_h2)
```

```
=== H2: Map distance predicts likeability ===
  Estimate  =  -1.1108
  SE        =   0.0160
  t         = -69.6313
  df        = 6723.9
  95% CI    = [-1.1421, -1.0795]
  p (one-sided, beta < 0) = 0.0000
  Supported : YES
```

3.1.3 H3

3.1.3.1 H3. Distances on the political map correspond to similarity ratings.

Similar to H2, we model similarity ratings between a pair of individuals, i and j, as a function of map distances:

$$s_{n,ij} = \beta_0 + \beta_1 \cdot d_{n,ij} + u_n + \epsilon_{n,ij} \ . \tag{3}$$

The prediction is $\beta_1 < 0$. Pairs placed farther apart on the map receive lower similarity ratings than pairs placed closer together.

```
df_h3 <- df_diff |>
  filter(wave %in% c(1, 2)) |>
  select(id, wave, pairwise_similarity, pixel_dist) |>
  drop_na()

m_h3  <- lmer(pairwise_similarity ~ pixel_dist + wave + (1 | id), data = df_h3, REML = TRUE)
ci_h3 <- confint(m_h3, parm = "pixel_dist", method = "Wald")
cf_h3 <- lmer_coef(m_h3, "pixel_dist")

print_one_sided("H3: Map distance predicts pairwise similarity",
                cf_h3$beta, cf_h3$se, cf_h3$t, cf_h3$df,
                direction = "negative", ci = ci_h3)
```

```
=== H3: Map distance predicts pairwise similarity ===
  Estimate  =  -0.7761
  SE        =   0.0066
  t         = -117.4882
  df        = 38233.2
  95% CI    = [-0.7890, -0.7631]
  p (one-sided, beta < 0) = 0.0000
  Supported : YES
```

3.1.4 H4

3.1.4.1 H4. The closest pair on the map is rated most similar and the farthest pair is rated least similar

Consider $(ij)^{min}$ the pair of individuals i and j with the smallest distance on participant n's political map and $(ij)^{max}$ the pair with the largest distance on the map. The similarity difference between these two pairs

$$\Delta s_{max} = s_{n,(ij)^{min}} - s_{n,(ij)^{max}} \ . \tag{4}$$

The prediction is $\Delta s_{max} > 0$ (standard t-test with $\alpha = 0.05$). The closest pair on the map receives a higher similarity rating than the farthest pair.

```r
df_h4_wide <- df_diff |>
  filter(wave %in% c(1, 2),
         minPixelDistance_pair_dummy == "True" | maxPixelDistance_pair_dummy == "True") |>
  select(id, wave, maxPixelDistance_pair_dummy, pairwise_similarity) |>
  pivot_wider(
    id_cols     = c(id, wave),
    names_from = maxPixelDistance_pair_dummy,
    values_from = pairwise_similarity
  ) |>
  rename(sim_min = "False", sim_max = "True") |>
  mutate(diff = sim_min - sim_max)

t_h4 <- t.test(df_h4_wide$diff, mu = 0)

cat("=== H4: Closest pair rated most similar ===\n")
cat(sprintf("  Mean(sim_min) = %.4f ± %.4f\n", mean(df_h4_wide$sim_min), sd(df_h4_wide$sim_min)))
cat(sprintf("  Mean(sim_max) = %.4f ± %.4f\n", mean(df_h4_wide$sim_max), sd(df_h4_wide$sim_max)))
cat(sprintf("  Mean diff     = %.4f\n",          mean(df_h4_wide$diff)))
cat(sprintf("  t             = %.4f\n",           t_h4$statistic))
cat(sprintf("  p (one-sided) = %.4f\n",           t_h4$p.value / 2))
cat(sprintf("  Supported     : %s\n\n",
            ifelse(t_h4$statistic > 0 && t_h4$p.value / 2 < 0.05, "YES", "NO")))
```

```
=== H4: Closest pair rated most similar ===
  Mean(sim_min) = 0.7113 ± 0.2688
  Mean(sim_max) = 0.2308 ± 0.2771
  Mean diff     = 0.4805
  t             = 55.3777
  p (one-sided) = 0.0000
  Supported     : YES
```

3.1.5 H5

3.1.5.1 H5. The inferred issue weights are robust across waves.

We test the reliability of our tool by comparing the similarity of issue weights, $\alpha_{n,q}$, of participants in wave 1 vs.~wave 2 (control condition) with the similarity of issue weights between participants.

$$\overline{\Delta\alpha}_{q_{\text{within}}} = \mathbb{E}\left[|\alpha_{n,q}(\text{wave }2) - \alpha_{n,q}(\text{wave }1)|\right] \qquad \text{and} \tag{5}$$

$$\overline{\Delta\alpha}_{q_{\text{between}}} = \mathbb{E}\left[|\alpha_{n,q}(\text{wave }2) - \alpha_{m,q}(\text{wave }1/\text{wave }2)|\right] \qquad \text{with } m \neq n \tag{6}$$

ERROR: It should be: $\overline{\Delta\alpha}_{q_{\text{within}}} - \overline{\Delta\alpha}_{q_{\text{between}}} < 0$
ADDON: $\mathbb{E}$ is the L1 (Manhattan) norm, i.e. $\mathbb{E}\left[|\alpha_{n,q}(\text{wave }2) - \alpha_{m,q}(\text{wave }1)|\right] = \sum_q \left[|\alpha_{n,q}(\text{wave }2) - \alpha_{m,q}(\text{wave }1)|\right]$.
ADDON: I compare only wave 2 alphas of pariticipant n to wave 1 alpahs of all other pariticipants m.

The prediction is that $\overline{\Delta\alpha}_{q_{\text{within}}} - \overline{\Delta\alpha}_{q_{\text{between}}} < 0$ (note: corrected error from preregistration) (one-sample t-test with $\alpha = 0.05$). Participants' issue weights in wave 2 are more similar to their own issue weights in the previous wave than to those of other participants in wave 1 (in the control condition).

```
k          <- "exp_alpha" # alternative: linear or logistic
alpha_cols <- paste0(k, "_", fitmode, "_", questions_sc)

ctrl_ids_h5 <- df_p |>
  filter(id %in% inds_bothwaves, wave == 2, treatment_wave2 == "0") |>
  pull(id)

df_h5 <- df_p |>
  filter(if_all(all_of(alpha_cols), Negate(is.na)), id %in% ctrl_ids_h5) |>
  select(id, wave, all_of(alpha_cols))

ids_both <- intersect(
  df_h5 |> filter(wave == 1) |> pull(id),
  df_h5 |> filter(wave == 2) |> pull(id)
)
df_h5 <- df_h5 |> filter(id %in% ids_both)

w1_mat <- df_h5 |> filter(wave == 1) |> arrange(id) |> select(all_of(alpha_cols)) |> as.matrix()
w2_mat <- df_h5 |> filter(wave == 2) |> arrange(id) |> select(all_of(alpha_cols)) |> as.matrix()
n      <- nrow(w1_mat)

within_means  <- rowMeans(abs(w2_mat - w1_mat))
between_means <- vapply(seq_len(n), function(i) {
  mean(rowMeans(abs(sweep(w2_mat[-i, , drop = FALSE], 2, w1_mat[i, ]))))
}, numeric(1))

delta    <- within_means - between_means
t_h5     <- t.test(delta, mu = 0, alternative = "two.sided")
cohens_d <- mean(delta) / sd(delta)

cat("=== H5: Issue weights reliable across waves ===\n")
cat(sprintf("  Within  (mean ± sd) = %.4f ± %.4f\n", mean(within_means), sd(within_means)))
cat(sprintf("  Between (mean ± sd) = %.4f ± %.4f\n", mean(between_means), sd(between_means)))
cat(sprintf("  Delta   (mean)      = %.4f\n",          mean(delta)))
cat(sprintf("  Cohen's d           = %.4f\n",          cohens_d))
cat(sprintf("  t                   = %.4f\n",           t_h5$statistic))
cat(sprintf("  p (two-sided)       = %.4f\n",           t_h5$p.value))
cat(sprintf("  p (one-sided)       = %.4f\n",           t_h5$p.value / 2))
cat(sprintf("  Supported           : %s\n\n",
            ifelse(t_h5$statistic < 0 && t_h5$p.value / 2 < 0.05, "YES", "NO")))
```

```
=== H5: Issue weights reliable across waves ===
  Within  (mean ± sd) = 0.1991 ± 0.0790
  Between (mean ± sd) = 0.2211 ± 0.0259
  Delta   (mean)      = -0.0220
  Cohen's d           = -0.3014
  t                   = -5.9894
  p (two-sided)       = 0.0000
  p (one-sided)       = 0.0000
  Supported           : YES
```

3.1.6 Exploration

3.1.6.1 Explorative Analyses.

In addition to the preregistered hypotheses above, we will explore how people rate the political mapping task against the pairwise similarity task in terms of difficulty and enjoyment, how satisfied they are with the political map, and how much time they needed for each task.

```
# (see 03_data-overview)
```

3.2 Aim 2

Aim 2: Identify patterns linked to (political) identity

In this section, we hypothesise that demographic and identity-related variables, including party preference, $party_n$, left-right placement, lr_n, age, age_n, or combinations of them moderate the relationship between belief differences and issue weights.

We have two different interprations of the following analysis: We can analyse the effect per issue separately (eq 2) or jointly across issues (eq 1).
Equation for Interpretation (1) for example for lr:

$$\alpha_{n,q} = beta_0 + \beta_{lr} \cdot lr + \sum_q \beta_q \cdot \mathbb{I}[issue = q] + \sum_q \beta_{lr \times q} lr \cdot \mathbb{I}[issue = q] + (u_n) + \beta_w wave + \epsilon_{n,q} \tag{7}$$

and equation for interpreatation (2) is for each issue q:

$$\alpha_{n,q} = beta_0 + \beta_{lr} \cdot lr + (u_n) + \beta_w wave + \epsilon_{n,q} \tag{8}$$

where u_n is the participant-level random intercept (gaussian distributed), included for weights defined as correlations (via lmer) but dropped for weights defined as the fitted weights with linear/exp kernels due to singular fits, where clustered standard errors are used instead

```r
run_H_joint <- function(k, predictor, label, use_lmer = TRUE, categorical = FALSE) {
  alpha_cols <- paste0(k, "_", questions_sc)
  df_h <- df_p |>
    filter(wave %in% c(1, 2)) |>
    select(id, wave, all_of(predictor), all_of(alpha_cols)) |>
    pivot_longer(
      cols         = all_of(alpha_cols),
      names_to     = "issue",
      names_prefix = paste0(k, "_"),
      values_to    = "alpha"
    ) |>
    filter(!is.na(.data[[predictor]]), !is.na(alpha)) |>
    mutate(
      issue = relevel(factor(issue), ref = "regulate_internet"),
      wave  = factor(wave)
    )

  if (categorical) {
    df_h <- df_h |>
      mutate(!!predictor := relevel(factor(.data[[predictor]]), ref = "No party"))
  }

  f_null <- alpha ~ issue + wave
  f_full <- reformulate(c("issue", paste0(predictor, " * issue"), "wave"), response = "alpha")

  if (use_lmer) {
    m_null <- lmer(update(f_null, . ~ . + (1 | id)), data = df_h, REML = FALSE)
    m_full <- lmer(update(f_full, . ~ . + (1 | id)), data = df_h, REML = FALSE)
    wt     <- anova(m_null, m_full)
    stat   <- wt$Chisq[2]; df_num <- wt$Df[2]; p <- wt$`Pr(>Chisq)`[2]
    print_fit_table("Null" = m_null, "Full" = m_full)
    tidy_print(m_full, mixed = TRUE)

    # Marginal and conditional R² for both models
    r2_null <- performance::r2(m_null)
    r2_full <- performance::r2(m_full)
    cat(sprintf(
      "\nR²  Null: marginal = %.3f, conditional = %.3f  |  Full: marginal = %.3f, conditional = %.3f\n",
      r2_null$R2_marginal, r2_null$R2_conditional,
      r2_full$R2_marginal, r2_full$R2_conditional
    ))

    print_lrt(sprintf("%s (issue weights: %s): %s × issue interaction", label, k, predictor),
              stat, df_num, p = p, type = "chi2")
  } else {
```

```r
    m_null <- lm(f_null, data = df_h)
    m_full <- lm(f_full, data = df_h)
    wt     <- anova(m_null, m_full)
    stat   <- wt$F[2]; df_num <- wt$Df[2]; df_den <- wt$Res.Df[2]; p <- wt$`Pr(>F)`[2]
    m_full_rob <- lm_robust(f_full, data = df_h, clusters = id, se_type = "CR2")

    cat(sprintf("\nR²  Null = %.3f  |  Full = %.3f\n",
                summary(m_null)$r.squared, summary(m_full)$r.squared))
    tidy_print(m_full_rob)
    print_lrt(sprintf("%s (issue weights: %s): %s × issue interaction", label, k, predictor),
              stat, df_num, df_den, p, type = "F")
  }
}


run_H_per_issue <- function(k, predictor, label, use_lmer = TRUE, categorical = FALSE) {
  cat(sprintf("\n-- %s per issue (issue weight: %s) --", label, k))

  purrr::walk(questions_sc, function(q) {
    df_q <- df_p |>
      filter(wave %in% c(1, 2)) |>
      select(id, wave, all_of(predictor), alpha_q = all_of(paste0(k,  "_", q))) |>
      filter(!is.na(.data[[predictor]]), !is.na(alpha_q)) |>
      mutate(wave = factor(wave))

    if (categorical) {
      df_q <- df_q |>
        mutate(!!predictor := relevel(factor(.data[[predictor]]), ref = "No party"))
    }

    f      <- reformulate(c(predictor, "wave"), response = "alpha_q")
    f_null <- alpha_q ~ wave

    if (use_lmer) {
      m      <- lmer(update(f, . ~ . + (1 | id)), data = df_q, REML = FALSE)
      m_null <- lmer(update(f_null, . ~ . + (1 | id)), data = df_q, REML = FALSE)
      wt     <- anova(m_null, m)
      p      <- wt$`Pr(>Chisq)`[2]; chi <- wt$Chisq[2]; df_lrt <- wt$Df[2]
      r2v    <- r2(m)
      cat(sprintf("\n%s  R^2=%.3f/%.3f  LRT chi^2(%d)=%.2f  p=%.4f %s ",
                  q, r2v$R2_marginal, r2v$R2_conditional,
                  df_lrt, chi, p, sig_stars(p)))
    } else {
      m      <- lm_robust(f, data = df_q, clusters = id, se_type = "CR2")
      m_lm   <- lm(f,      data = df_q)
      m_null <- lm(f_null, data = df_q)
      wt     <- anova(m_null, m_lm)
      p      <- wt$`Pr(>F)`[2]; f_stat <- wt$F[2]
      cat(sprintf("\n%s  R^2=%.3f  F(%d,%d)=%.2f  p=%.4f %s ",
                  q, summary(m_lm)$r.squared,
                  wt$Df[2], wt$Res.Df[2], f_stat, p, sig_stars(p)))
    }
  })
}
```

3.2.1 H6

3.2.1.1 H6. Party preference predicts issue weights.

We model the (normalised) issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \sum_r \beta_{r,q} \cdot \mathbb{I}[\text{party}_n = r] + u_q + \epsilon_{n,q} \tag{9}$$

where $\mathbb{I}[\text{party}_n = r]$ are dummy variables for each party r (with one party as reference category), $\beta_{r,q}$ is the effect of party r on weight assigned to issue q, u_q is the average deviation of the weight for issue q, and $\epsilon_{n,q}$ is the individual- and issue-specific error.

The prediction is $\beta_{r,q} \neq 0$ and $\beta_{r1,q} \neq \beta_{r2,q}$ for at least some parties $r1$ and $r2$. Participants affiliated with different parties differ in how strongly they weigh different political issues when mapping individuals.

Different interprations: Per Issue (1) or Jointly across issues (2)?
Equation for Interpretation (1):

$$\alpha_{n,q} = beta_0 + \sum_r \beta_r \cdot \mathbb{I}[\text{party}_n = r] + \sum_q \beta_q q(+u_n) + \sum_q \beta_{r,q} \mathbb{I}[\text{party}_n = r] \cdot q + (u_n) + v_q + \epsilon_{n,q} \tag{10}$$

and equation for interpreatation (2) is for each issue q:

$$\alpha_{n,q} = beta_0 + \sum_r \beta_r \cdot \mathbb{I}[\text{party}_n = r] + (u_n) + v_q + \epsilon_{n,q} \tag{11}$$

where u_n is removed if the alphas are constrained (in the case of linear/exp fit).

Interpretation 1: Do parties differ across issues?

For exponential/linear weights (constrained):

```
# H6 - party
run_H_joint    (paste0("exp_alpha_", fitmode), "party_close", "H6", use_lmer = FALSE, categorical = TRUE)
run_H_joint    (paste0("corrP_alpha_", fitmode),  "party_close", "H6", use_lmer = TRUE,  categorical = TRUE)
```

```
R²  Null = 0.008  |  Full = 0.015
                                               term estimate
1                                       (Intercept)    0.142
2                                 issueclimate_concern    0.039
3                                   issueeast_germans    0.021
4                                 issueecon_inequality    0.045
5                                   issuegay_marriage   -0.004
6                        issuerights_indep_integration    0.047
7                                     party_closeAfD   -0.026
8                                     party_closeBSW   -0.055
9                                 party_closeCDU/CSU   -0.001
10                                    party_closeFDP   -0.007
11                            party_closeGreen Party   -0.020
12                             party_closeLeft Party   -0.011
13                            party_closeOther party   -0.020
14                  party_closeRefuse to say/No answer    0.106
15                                    party_closeSPD    0.003
16                                             wave2    0.000
17                  issueclimate_concern:party_closeAfD    0.027
18                    issueeast_germans:party_closeAfD    0.029
19                  issueecon_inequality:party_closeAfD    0.013
20                    issuegay_marriage:party_closeAfD    0.025
21         issuerights_indep_integration:party_closeAfD    0.059
22                  issueclimate_concern:party_closeBSW    0.073
23                    issueeast_germans:party_closeBSW    0.106
24                  issueecon_inequality:party_closeBSW    0.038
25                    issuegay_marriage:party_closeBSW    0.084
26         issuerights_indep_integration:party_closeBSW    0.026
27              issueclimate_concern:party_closeCDU/CSU    0.002
28                issueeast_germans:party_closeCDU/CSU    0.010
29              issueecon_inequality:party_closeCDU/CSU   -0.046
30                issuegay_marriage:party_closeCDU/CSU    0.036
31     issuerights_indep_integration:party_closeCDU/CSU    0.003
32                  issueclimate_concern:party_closeFDP   -0.009
33                    issueeast_germans:party_closeFDP    0.012
34                  issueecon_inequality:party_closeFDP   -0.028
35                    issuegay_marriage:party_closeFDP    0.020
36         issuerights_indep_integration:party_closeFDP    0.045
37          issueclimate_concern:party_closeGreen Party    0.047
38            issueeast_germans:party_closeGreen Party    0.024
39          issueecon_inequality:party_closeGreen Party   -0.032
```

```
40                        issuegay_marriage:party_closeGreen Party    0.062
41            issuerights_indep_integration:party_closeGreen Party    0.019
42                      issueclimate_concern:party_closeLeft Party    0.060
43                         issueeast_germans:party_closeLeft Party   -0.023
44                      issueecon_inequality:party_closeLeft Party   -0.021
45                         issuegay_marriage:party_closeLeft Party    0.038
46             issuerights_indep_integration:party_closeLeft Party    0.010
47                     issueclimate_concern:party_closeOther party    0.069
48                        issueeast_germans:party_closeOther party    0.017
49                     issueecon_inequality:party_closeOther party   -0.020
50                        issuegay_marriage:party_closeOther party    0.071
51            issuerights_indep_integration:party_closeOther party   -0.016
52         issueclimate_concern:party_closeRefuse to say/No answer   -0.176
53            issueeast_germans:party_closeRefuse to say/No answer   -0.066
54         issueecon_inequality:party_closeRefuse to say/No answer   -0.121
55            issuegay_marriage:party_closeRefuse to say/No answer   -0.065
56 issuerights_indep_integration:party_closeRefuse to say/No answer   -0.211
57                               issueclimate_concern:party_closeSPD    0.003
58                                  issueeast_germans:party_closeSPD    0.000
59                               issueecon_inequality:party_closeSPD   -0.022
60                                  issuegay_marriage:party_closeSPD    0.015
61                      issuerights_indep_integration:party_closeSPD   -0.017
   std.error p.value
1      0.011   0.000
2      0.019   0.042
3      0.018   0.235
4      0.021   0.035
5      0.017   0.793
6      0.018   0.009
7      0.015   0.086
8      0.020   0.008
9      0.015   0.954
10     0.023   0.776
11     0.016   0.195
12     0.017   0.525
13     0.031   0.525
14     0.065   0.119
15     0.016   0.836
16     0.000   0.424
17     0.026   0.287
18     0.025   0.258
19     0.027   0.635
20     0.023   0.283
21     0.026   0.025
22     0.041   0.077
23     0.044   0.018
24     0.038   0.316
25     0.042   0.047
26     0.037   0.485
27     0.026   0.949
28     0.025   0.685
29     0.026   0.085
30     0.024   0.134
31     0.025   0.894
32     0.040   0.823
33     0.039   0.762
34     0.036   0.440
35     0.036   0.584
36     0.041   0.282
37     0.027   0.086
38     0.024   0.323
39     0.027   0.239
40     0.025   0.013
41     0.025   0.445
```

```
42     0.031   0.055
43     0.027   0.382
44     0.030   0.495
45     0.027   0.170
46     0.031   0.745
47     0.046   0.141
48     0.060   0.784
49     0.049   0.679
50     0.056   0.213
51     0.054   0.767
52     0.076   0.035
53     0.098   0.513
54     0.089   0.193
55     0.105   0.544
56     0.085   0.024
57     0.028   0.917
58     0.026   0.992
59     0.028   0.439
60     0.025   0.557
61     0.026   0.515
=== H6 (issue weights: exp_alpha_fitAllDots): party_close × issue interaction ===
  F(54, 12713) = 1.519,  p = 0.0084  **

 Model     AIC     BIC logLik npar delta_AIC delta_BIC
  Null -2358.8 -2291.7 1188.4    7     284.9       0.0
  Full -2643.7 -2174.1 1384.9   61       0.0     117.6
# A tibble: 61 x 4
   term                                                         estimate
   <chr>                                                           <dbl>
 1 (Intercept)                                                     0.201
 2 issueclimate_concern                                            0.094
 3 issueeast_germans                                               0.018
 4 issueecon_inequality                                            0.037
 5 issuegay_marriage                                               0.06
 6 issuerights_indep_integration                                   0.114
 7 party_closeAfD                                                  0.04
 8 party_closeBSW                                                 -0.022
 9 party_closeCDU/CSU                                             -0.039
10 party_closeFDP                                                 -0.06
11 party_closeGreen Party                                          0.044
12 party_closeLeft Party                                          -0.003
13 party_closeOther party                                         -0.084
14 party_closeRefuse to say/No answer                              0.138
15 party_closeSPD                                                 -0.007
16 wave2                                                           0.006
17 issueclimate_concern:party_closeAfD                             0.011
18 issueeast_germans:party_closeAfD                               -0.011
19 issueecon_inequality:party_closeAfD                            -0.014
20 issuegay_marriage:party_closeAfD                               -0.055
21 issuerights_indep_integration:party_closeAfD                    0.027
22 issueclimate_concern:party_closeBSW                             0.042
23 issueeast_germans:party_closeBSW                                0.023
24 issueecon_inequality:party_closeBSW                             0.027
25 issuegay_marriage:party_closeBSW                                0.02
26 issuerights_indep_integration:party_closeBSW                   -0.006
27 issueclimate_concern:party_closeCDU/CSU                        -0.023
28 issueeast_germans:party_closeCDU/CSU                            0.029
29 issueecon_inequality:party_closeCDU/CSU                        -0.01
30 issuegay_marriage:party_closeCDU/CSU                            0.007
31 issuerights_indep_integration:party_closeCDU/CSU               -0.01
32 issueclimate_concern:party_closeFDP                             0.035
33 issueeast_germans:party_closeFDP                                0.084
34 issueecon_inequality:party_closeFDP                             0.041
35 issuegay_marriage:party_closeFDP                                0.047
```

```
36 issuerights_indep_integration:party_closeFDP                          0.077
37 issueclimate_concern:party_closeGreen Party                           0.092
38 issueeast_germans:party_closeGreen Party                             -0.018
39 issueecon_inequality:party_closeGreen Party                           0.026
40 issuegay_marriage:party_closeGreen Party                              0.117
41 issuerights_indep_integration:party_closeGreen Party                  0.037
42 issueclimate_concern:party_closeLeft Party                            0.137
43 issueeast_germans:party_closeLeft Party                              -0.036
44 issueecon_inequality:party_closeLeft Party                            0.109
45 issuegay_marriage:party_closeLeft Party                               0.149
46 issuerights_indep_integration:party_closeLeft Party                   0.095
47 issueclimate_concern:party_closeOther party                           0.062
48 issueeast_germans:party_closeOther party                              0.004
49 issueecon_inequality:party_closeOther party                           0.048
50 issuegay_marriage:party_closeOther party                              0.041
51 issuerights_indep_integration:party_closeOther party                  0.025
52 issueclimate_concern:party_closeRefuse to say/No answer              -0.112
53 issueeast_germans:party_closeRefuse to say/No answer                 -0.076
54 issueecon_inequality:party_closeRefuse to say/No answer              -0.035
55 issuegay_marriage:party_closeRefuse to say/No answer                 -0.062
56 issuerights_indep_integration:party_closeRefuse to say/No answer   -0.118
57 issueclimate_concern:party_closeSPD                                   0.012
58 issueeast_germans:party_closeSPD                                     -0.02
59 issueecon_inequality:party_closeSPD                                   0.016
60 issuegay_marriage:party_closeSPD                                      0.031
61 issuerights_indep_integration:party_closeSPD                         -0.052
   std.error p.value
       <dbl>   <dbl>
 1     0.014   0
 2     0.015   0
 3     0.015   0.229
 4     0.015   0.016
 5     0.015   0
 6     0.015   0
 7     0.019   0.037
 8     0.032   0.483
 9     0.018   0.035
10     0.032   0.061
11     0.022   0.042
12     0.025   0.918
13     0.038   0.029
14     0.054   0.011
15     0.02    0.731
16     0.004   0.099
17     0.021   0.616
18     0.021   0.609
19     0.021   0.498
20     0.021   0.009
21     0.021   0.202
22     0.036   0.237
23     0.036   0.523
24     0.036   0.445
25     0.036   0.579
26     0.036   0.857
27     0.02    0.255
28     0.02    0.147
29     0.02    0.633
30     0.02    0.733
31     0.02    0.604
32     0.036   0.326
33     0.036   0.02
34     0.036   0.257
35     0.036   0.187
36     0.036   0.032
```

```
37      0.022   0
38      0.022   0.425
39      0.022   0.249
40      0.022   0
41      0.022   0.096
42      0.026   0
43      0.026   0.164
44      0.026   0
45      0.026   0
46      0.026   0
47      0.044   0.161
48      0.044   0.922
49      0.044   0.28
50      0.044   0.353
51      0.044   0.569
52      0.064   0.08
53      0.064   0.235
54      0.064   0.589
55      0.064   0.335
56      0.064   0.065
57      0.022   0.582
58      0.022   0.345
59      0.022   0.452
60      0.022   0.157
61      0.022   0.016

R²  Null: marginal = 0.032, conditional = 0.439  |  Full: marginal = 0.073, conditional = 0.452
=== H6 (issue weights: corrP_alpha_fitAllDots): party_close × issue interaction ===
  chi^2(54) = 392.980,  p = 0.0000  ***
```

Interpretation 2: Within each question, do parties differ?

```
run_H_per_issue(paste0("exp_alpha_", fitmode),    "party_close", "H6", use_lmer = FALSE, categorical = TRUE)
run_H_per_issue(paste0("corrP_alpha_", fitmode),  "party_close", "H6", use_lmer = TRUE,  categorical = TRUE)
```

```
-- H6 per issue (issue weight: exp_alpha_fitAllDots) --
climate_concern  R^2=0.006  F(9,2118)=1.43  p=0.1696 ns
gay_marriage  R^2=0.006  F(9,2118)=1.30  p=0.2309 ns
rights_indep_integration  R^2=0.008  F(9,2118)=1.81  p=0.0620 .
econ_inequality  R^2=0.007  F(9,2118)=1.71  p=0.0816 .
regulate_internet  R^2=0.008  F(9,2118)=1.86  p=0.0543 .
east_germans  R^2=0.006  F(9,2118)=1.03  p=0.4137 ns
-- H6 per issue (issue weight: corrP_alpha_fitAllDots) --
climate_concern  R^2=0.062/0.557  LRT chi^2(9)=100.48  p=0.0000 ***
gay_marriage  R^2=0.072/0.535  LRT chi^2(9)=118.38  p=0.0000 ***
rights_indep_integration  R^2=0.049/0.540  LRT chi^2(9)=81.67  p=0.0000 ***
econ_inequality  R^2=0.035/0.475  LRT chi^2(9)=58.79  p=0.0000 ***
regulate_internet  R^2=0.026/0.409  LRT chi^2(9)=45.72  p=0.0000 ***
east_germans  R^2=0.016/0.429  LRT chi^2(9)=27.01  p=0.0014 **
```

3.2.2 H7

3.2.2.1 H7. Left-Right placement predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot lr_n + u_q + \epsilon_{n,q} \tag{12}$$

with continuous variable lr_n between `left' ($-5$) and`right' (5). We estimate this model jointly across all six issues. % beta_0 is average. Because alpha is normalised: beta_0 = 1/6 % lr_n is explanatory variable % u_q is average deviation of the weight for issue q % eps_{nq} is individual specific error for this question and this individual.

The prediction is $\beta_q \neq 0$. Left and right participants differ in how strongly they weigh different political issues when mapping individuals. %We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```r
# H7 - lr
run_H_joint    (paste0("exp_alpha_", fitmode), "lr", "H7", use_lmer = FALSE)
run_H_joint    (paste0("corrP_alpha_", fitmode),  "lr", "H7", use_lmer = TRUE)
```

```
R²  Null = 0.008  |  Full = 0.010
                               term estimate std.error p.value
1                       (Intercept)    0.127     0.010   0.000
2              issueclimate_concern    0.095     0.018   0.000
3                 issueeast_germans    0.015     0.016   0.346
4              issueecon_inequality    0.025     0.016   0.116
5                 issuegay_marriage    0.061     0.017   0.000
6     issuerights_indep_integration    0.043     0.018   0.016
7                                lr    0.013     0.018   0.491
8                             wave2    0.000     0.000   0.428
9           issueclimate_concern:lr   -0.077     0.035   0.028
10             issueeast_germans:lr    0.036     0.032   0.261
11          issueecon_inequality:lr    0.005     0.032   0.883
12             issuegay_marriage:lr   -0.073     0.032   0.023
13 issuerights_indep_integration:lr    0.033     0.034   0.326
=== H7 (issue weights: exp_alpha_fitAllDots): lr × issue interaction ===
  F(6, 12761) = 4.085,  p = 0.0004  ***

 Model     AIC     BIC logLik npar delta_AIC delta_BIC
  Null -2358.8 -2291.7 1188.4    7     186.6     141.9
  Full -2545.4 -2433.6 1287.7   13       0.0       0.0
# A tibble: 13 x 4
   term                             estimate std.error p.value
   <chr>                               <dbl>     <dbl>   <dbl>
 1 (Intercept)                         0.201     0.015   0
 2 issueclimate_concern                0.221     0.015   0
 3 issueeast_germans                   0         0.015   0.989
 4 issueecon_inequality                0.124     0.015   0
 5 issuegay_marriage                   0.218     0.015   0
 6 issuerights_indep_integration       0.154     0.015   0
 7 lr                                 -0.002     0.029   0.945
 8 wave2                               0.008     0.004   0.023
 9 issueclimate_concern:lr            -0.209     0.029   0
10 issueeast_germans:lr                0.035     0.029   0.23
11 issueecon_inequality:lr            -0.152     0.029   0
12 issuegay_marriage:lr               -0.269     0.029   0
13 issuerights_indep_integration:lr   -0.063     0.029   0.03

R²  Null: marginal = 0.032, conditional = 0.439  |  Full: marginal = 0.048, conditional = 0.445
=== H7 (issue weights: corrP_alpha_fitAllDots): lr × issue interaction ===
  chi^2(6) = 198.683,  p = 0.0000  ***
```

And again, per issue:

```r
run_H_per_issue(paste0("exp_alpha_", fitmode),    "lr", "H7", use_lmer = FALSE)
run_H_per_issue(paste0("corrP_alpha_", fitmode),  "lr", "H7", use_lmer = TRUE)
```

```
-- H7 per issue (issue weight: exp_alpha_fitAllDots) --
climate_concern  R^2=0.003  F(1,2126)=7.23  p=0.0072 **
gay_marriage  R^2=0.004  F(1,2126)=6.92  p=0.0086 **
rights_indep_integration  R^2=0.002  F(1,2126)=3.55  p=0.0595 .
econ_inequality  R^2=0.000  F(1,2126)=0.62  p=0.4317 ns
regulate_internet  R^2=0.001  F(1,2126)=0.41  p=0.5244 ns
east_germans  R^2=0.004  F(1,2126)=4.39  p=0.0362 *
-- H7 per issue (issue weight: corrP_alpha_fitAllDots) --
climate_concern  R^2=0.031/0.567  LRT chi^2(1)=50.01  p=0.0000 ***
gay_marriage  R^2=0.047/0.530  LRT chi^2(1)=77.36  p=0.0000 ***
rights_indep_integration  R^2=0.005/0.547  LRT chi^2(1)=7.66  p=0.0056 **
econ_inequality  R^2=0.021/0.479  LRT chi^2(1)=33.64  p=0.0000 ***
```

```
regulate_internet  R^2=0.001/0.404  LRT chi^2(1)=0.52  p=0.4728 ns
east_germans  R^2=0.000/0.427  LRT chi^2(1)=0.00  p=0.9496 ns
```

3.2.3 H8

3.2.3.1 H8. Age predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot age_n + u_q + \epsilon_{n,q} \tag{13}$$

The prediction is $\beta_q \neq 0$. Older and younger participants differ in how strongly they weigh different political issues when mapping individuals.

```r
run_H_joint    (paste0("exp_alpha_", fitmode), "age", "H8", use_lmer = FALSE)
run_H_joint    (paste0("corrP_alpha_", fitmode),  "age", "H8", use_lmer = TRUE)
```

```
R²  Null = 0.008  |  Full = 0.010
                                term estimate std.error p.value
1                        (Intercept)    0.125     0.015   0.000
2                 issueclimate_concern    0.024     0.026   0.346
3                   issueeast_germans    0.009     0.024   0.705
4                issueecon_inequality    0.052     0.025   0.040
5                   issuegay_marriage    0.069     0.026   0.009
6       issuerights_indep_integration    0.094     0.026   0.000
7                                 age    0.000     0.000   0.579
8                               wave2    0.000     0.000   0.424
9             issueclimate_concern:age    0.001     0.001   0.170
10              issueeast_germans:age    0.000     0.000   0.311
11           issueecon_inequality:age   -0.001     0.000   0.290
12              issuegay_marriage:age   -0.001     0.001   0.073
13 issuerights_indep_integration:age   -0.001     0.001   0.143
=== H8 (issue weights: exp_alpha_fitAllDots): age × issue interaction ===
  F(6, 12659) = 3.491,  p = 0.0019  **

 Model     AIC   BIC logLik npar delta_AIC delta_BIC
  Null -2319.0 -2252 1168.5    7      27.7         0
  Full -2346.7 -2235 1188.3   13       0.0        17
# A tibble: 13 x 4
   term                              estimate std.error p.value
   <chr>                                <dbl>     <dbl>   <dbl>
 1 (Intercept)                          0.159     0.022   0
 2 issueclimate_concern                 0.154     0.021   0
 3 issueeast_germans                    0.005     0.021   0.807
 4 issueecon_inequality                 0.112     0.021   0
 5 issuegay_marriage                    0.159     0.021   0
 6 issuerights_indep_integration        0.209     0.021   0
 7 age                                  0.001     0       0.051
 8 wave2                                0.008     0.004   0.041
 9 issueclimate_concern:age            -0.001     0       0.091
10 issueeast_germans:age                0         0       0.553
11 issueecon_inequality:age            -0.001     0       0.002
12 issuegay_marriage:age               -0.001     0       0
13 issuerights_indep_integration:age   -0.002     0       0

R²  Null: marginal = 0.032, conditional = 0.440  |  Full: marginal = 0.034, conditional = 0.442
=== H8 (issue weights: corrP_alpha_fitAllDots): age × issue interaction ===
  chi^2(6) = 39.647,  p = 0.0000  ***
```

And again, per issue:

```r
run_H_per_issue(paste0("exp_alpha_", fitmode),    "age", "H8", use_lmer = FALSE)
run_H_per_issue(paste0("corrP_alpha_", fitmode),  "age", "H8", use_lmer = TRUE)
```

```
-- H8 per issue (issue weight: exp_alpha_fitAllDots) --
climate_concern  R^2=0.003  F(1,2109)=6.40  p=0.0115 *
gay_marriage  R^2=0.003  F(1,2109)=5.26  p=0.0219 *
rights_indep_integration  R^2=0.002  F(1,2109)=2.93  p=0.0873 .
econ_inequality  R^2=0.001  F(1,2109)=1.20  p=0.2726 ns
regulate_internet  R^2=0.001  F(1,2109)=0.31  p=0.5748 ns
east_germans  R^2=0.004  F(1,2109)=3.98  p=0.0462 *
-- H8 per issue (issue weight: corrP_alpha_fitAllDots) --
climate_concern  R^2=0.001/0.570  LRT chi^2(1)=0.01  p=0.9344 ns
gay_marriage  R^2=0.001/0.537  LRT chi^2(1)=1.76  p=0.1844 ns
rights_indep_integration  R^2=0.003/0.549  LRT chi^2(1)=3.70  p=0.0545 .
econ_inequality  R^2=0.001/0.484  LRT chi^2(1)=0.74  p=0.3901 ns
regulate_internet  R^2=0.003/0.406  LRT chi^2(1)=4.11  p=0.0426 *
east_germans  R^2=0.005/0.427  LRT chi^2(1)=7.42  p=0.0064 **
```

3.2.4 Exploration

3.2.4.1 Explorative Analyses.

We also model **H6** to **H8** with different combinations of party preference, left-right placement, and age of the participants to model participants issue weights $\alpha_{n,q}$. In addition, we model how political interest, pi_n, pf_n, interacts with the explanatory variables above.

```r
print(" exploration: use all predictors for issue weight prediction")

k = paste0("exp_alpha_", fitmode)

use_lmer = FALSE
label = "demographics"
cat(sprintf("\n-- %s per issue (issue weight: %s) --", label, k))
predictor = c("age", "lr", "party_close", "gender")
models = purrr::map(questions_sc, function(q) {
  alpha_col <- paste0(k, "_", q)
  df_q <- df_p |>
    filter(wave %in% c(1, 2)) |>
    select(id, wave, all_of(predictor), alpha_q = all_of(alpha_col)) |>
    filter(gender %in% c("m", "f")) |>
    filter(!is.na(age), !is.na(lr), !is.na(party_close), !is.na(alpha_q), !is.na(gender)) |>
    mutate(wave = factor(wave))|>
    mutate(gender = factor(gender))|>
    mutate(age_z = scale(age))|>
    mutate(party_close = relevel(factor(party_close), ref = "No party"))


  f      <- alpha_q ~ wave + age_z * lr * party_close  * gender
  f_null <- alpha_q ~ wave

  if (use_lmer) {
    m      <- lmer(update(f, . ~ . + (1 | id)), data = df_q, REML = FALSE)
    m_null <- lmer(update(f_null, . ~ . + (1 | id)), data = df_q, REML = FALSE)
    wt     <- anova(m_null, m)
    p      <- wt$`Pr(>Chisq)`[2]; chi <- wt$Chisq[2]; df_lrt <- wt$Df[2]
    r2v    <- r2(m)
    cat(sprintf("\n%s  R^2=%.3f/%.3f  LRT chi^2(%d)=%.2f  p=%.4f %s ",
                q, r2v$R2_marginal, r2v$R2_conditional,
                df_lrt, chi, p, sig_stars(p)))
  } else {
    m      <- lm_robust(f, data = df_q, clusters = id, se_type = "CR2")
    m   <- lm(f,      data = df_q)
    m_null <- lm(f_null, data = df_q)
    wt     <- anova(m_null, m)
    p      <- wt$`Pr(>F)`[2]; f_stat <- wt$F[2]
    cat(sprintf("\n%s  R^2=%.3f  F(%d,%d)=%.2f  p=%.4f %s ",
                q, summary(m)$r.squared,
                wt$Df[2], wt$Res.Df[2], f_stat, p, sig_stars(p)))
```

```
  }
  m
})
cat (questions_sc[[1]])
x <- models |> purrr::pluck(1) |> tidy_print()
```

```
[1] " exploration: use all predictors for issue weight prediction"

-- demographics per issue (issue weight: exp_alpha_fitAllDots) --
climate_concern  R^2=0.045  F(79,2024)=1.22  p=0.0969 .
gay_marriage  R^2=0.044  F(79,2024)=1.16  p=0.1667 ns
rights_indep_integration  R^2=0.045  F(79,2024)=1.21  p=0.1036 ns
econ_inequality  R^2=0.046  F(79,2024)=1.22  p=0.0941 .
regulate_internet  R^2=0.050  F(79,2024)=1.33  p=0.0308 *
east_germans  R^2=0.059  F(79,2024)=1.56  p=0.0014 ** climate_concern# A tibble: 81 x 4
   term                                                 estimate std.error
   <chr>                                                   <dbl>     <dbl>
 1 (Intercept)                                             0.172     0.068
 2 wave2                                                   0.002     0.01
 3 age_z                                                  -0.012     0.072
 4 lr                                                     -0.022     0.136
 5 party_closeAfD                                         -0.03      0.106
 6 party_closeBSW                                         -0.215     0.12
 7 party_closeCDU/CSU                                     -0.032     0.103
 8 party_closeFDP                                          0.254     0.371
 9 party_closeGreen Party                                  0.072     0.088
10 party_closeLeft Party                                   0.097     0.082
11 party_closeOther party                                  0.354     0.231
12 party_closeRefuse to say/No answer                     -0.084     0.44
13 party_closeSPD                                          0.085     0.086
14 genderm                                                 0.156     0.097
15 age_z:lr                                                0.036     0.145
16 age_z:party_closeAfD                                   -0.117     0.112
17 age_z:party_closeBSW                                   -0.01      0.11
18 age_z:party_closeCDU/CSU                               -0.057     0.109
19 age_z:party_closeFDP                                    0.056     0.324
20 age_z:party_closeGreen Party                            0.025     0.085
21 age_z:party_closeLeft Party                             0.038     0.08
22 age_z:party_closeOther party                            0.419     0.227
23 age_z:party_closeRefuse to say/No answer              -0.389     0.636
24 age_z:party_closeSPD                                    0.093     0.086
25 lr:party_closeAfD                                       0.083     0.183
26 lr:party_closeBSW                                       0.569     0.245
27 lr:party_closeCDU/CSU                                   0.045     0.191
28 lr:party_closeFDP                                      -0.437     0.64
29 lr:party_closeGreen Party                              -0.095     0.203
30 lr:party_closeLeft Party                               -0.098     0.213
31 lr:party_closeOther party                              -0.559     0.468
32 lr:party_closeRefuse to say/No answer                   0.04      0.952
33 lr:party_closeSPD                                      -0.177     0.181
34 age_z:genderm                                           0.022     0.103
35 lr:genderm                                             -0.207     0.186
36 party_closeAfD:genderm                                 -0.203     0.146
37 party_closeBSW:genderm                                  0.162     0.164
38 party_closeCDU/CSU:genderm                             -0.008     0.137
39 party_closeFDP:genderm                                 -0.393     0.407
40 party_closeGreen Party:genderm                         -0.125     0.122
41 party_closeLeft Party:genderm                          -0.176     0.119
42 party_closeOther party:genderm                         -0.34      0.263
43 party_closeRefuse to say/No answer:genderm              0.348     0.548
44 party_closeSPD:genderm                                 -0.174     0.121
45 age_z:lr:party_closeAfD                                 0.208     0.194
46 age_z:lr:party_closeBSW                                 0.012     0.246
```

```
47 age_z:lr:party_closeCDU/CSU                         0.107     0.206
48 age_z:lr:party_closeFDP                             0.098     0.566
49 age_z:lr:party_closeGreen Party                     0.033     0.196
50 age_z:lr:party_closeLeft Party                     -0.001     0.194
51 age_z:lr:party_closeOther party                    -0.711     0.392
52 age_z:lr:party_closeRefuse to say/No answer         0.766     1.31
53 age_z:lr:party_closeSPD                            -0.198     0.188
54 age_z:lr:genderm                                   -0.022     0.199
55 age_z:party_closeAfD:genderm                        0.102     0.154
56 age_z:party_closeBSW:genderm                       -0.074     0.163
57 age_z:party_closeCDU/CSU:genderm                    0.197     0.145
58 age_z:party_closeFDP:genderm                       -0.245     0.368
59 age_z:party_closeGreen Party:genderm                0.061     0.127
60 age_z:party_closeLeft Party:genderm                 0.044     0.118
61 age_z:party_closeOther party:genderm               -0.233     0.275
62 age_z:party_closeRefuse to say/No answer:genderm    0.745     0.676
63 age_z:party_closeSPD:genderm                       -0.08      0.123
64 lr:party_closeAfD:genderm                           0.27      0.244
65 lr:party_closeBSW:genderm                          -0.513     0.353
66 lr:party_closeCDU/CSU:genderm                       0.027     0.248
67 lr:party_closeFDP:genderm                           0.664     0.701
68 lr:party_closeGreen Party:genderm                   0.13      0.268
69 lr:party_closeLeft Party:genderm                    0.225     0.325
70 lr:party_closeOther party:genderm                   0.509     0.532
71 lr:party_closeRefuse to say/No answer:genderm      -0.923     1.33
72 lr:party_closeSPD:genderm                           0.289     0.244
73 age_z:lr:party_closeAfD:genderm                    -0.19      0.259
74 age_z:lr:party_closeBSW:genderm                    -0.017     0.346
75 age_z:lr:party_closeCDU/CSU:genderm                -0.372     0.266
76 age_z:lr:party_closeFDP:genderm                     0.144     0.636
77 age_z:lr:party_closeGreen Party:genderm            -0.328     0.274
78 age_z:lr:party_closeLeft Party:genderm             -0.317     0.295
79 age_z:lr:party_closeOther party:genderm             0.404     0.495
80 age_z:lr:party_closeRefuse to say/No answer:genderm   -1.59      1.51
81 age_z:lr:party_closeSPD:genderm                     0.174     0.249
   p.value
     <dbl>
 1   0.012
 2   0.808
 3   0.863
 4   0.871
 5   0.775
 6   0.074
 7   0.753
 8   0.493
 9   0.414
10   0.236
11   0.125
12   0.848
13   0.32
14   0.109
15   0.805
16   0.297
17   0.925
18   0.599
19   0.863
20   0.765
21   0.633
22   0.066
23   0.541
24   0.281
25   0.65
26   0.02
27   0.814
```

```
28   0.495
29   0.642
30   0.646
31   0.232
32   0.966
33   0.326
34   0.833
35   0.267
36   0.167
37   0.322
38   0.951
39   0.334
40   0.308
41   0.139
42   0.196
43   0.526
44   0.149
45   0.285
46   0.96
47   0.603
48   0.863
49   0.866
50   0.994
51   0.07
52   0.558
53   0.294
54   0.913
55   0.507
56   0.65
57   0.173
58   0.506
59   0.631
60   0.71
61   0.398
62   0.27
63   0.519
64   0.269
65   0.147
66   0.915
67   0.344
68   0.627
69   0.489
70   0.339
71   0.489
72   0.237
73   0.462
74   0.962
75   0.162
76   0.82
77   0.231
78   0.283
79   0.415
80   0.291
81   0.485
```

3.2.5 Exploration 2: Can we predict the participant-reported issue importance based on party, lr, and age any better?

```
run_H_joint    ("w", "party_close", "H6 Weights", use_lmer = TRUE)
```

```
 Model    AIC    BIC  logLik npar delta_AIC delta_BIC
  Null 2457.1 2524.2 -1219.6    7     697.9     295.3
  Full 1759.2 2228.9  -816.6   61       0.0       0.0
# A tibble: 61 x 4
```

```
   term                                                              estimate
   <chr>                                                                <dbl>
 1 (Intercept)                                                          0.534
 2 issueclimate_concern                                                -0.142
 3 issueeast_germans                                                    0.008
 4 issueecon_inequality                                                 0.047
 5 issuegay_marriage                                                   -0.087
 6 issuerights_indep_integration                                        0.013
 7 party_closeBSW                                                       0.01
 8 party_closeCDU/CSU                                                   0.049
 9 party_closeFDP                                                      -0.018
10 party_closeGreen Party                                               0.022
11 party_closeLeft Party                                               -0.03
12 party_closeNo party                                                 -0.05
13 party_closeOther party                                              -0.064
14 party_closeRefuse to say/No answer                                  -0.034
15 party_closeSPD                                                       0.096
16 wave2                                                                0.001
17 issueclimate_concern:party_closeBSW                                  0.133
18 issueeast_germans:party_closeBSW                                     0.004
19 issueecon_inequality:party_closeBSW                                  0.104
20 issuegay_marriage:party_closeBSW                                     0.074
21 issuerights_indep_integration:party_closeBSW                         0.061
22 issueclimate_concern:party_closeCDU/CSU                              0.177
23 issueeast_germans:party_closeCDU/CSU                                -0.096
24 issueecon_inequality:party_closeCDU/CSU                             -0.012
25 issuegay_marriage:party_closeCDU/CSU                                 0.087
26 issuerights_indep_integration:party_closeCDU/CSU                    -0.005
27 issueclimate_concern:party_closeFDP                                  0.155
28 issueeast_germans:party_closeFDP                                    -0.036
29 issueecon_inequality:party_closeFDP                                 -0.042
30 issuegay_marriage:party_closeFDP                                     0.106
31 issuerights_indep_integration:party_closeFDP                         0.062
32 issueclimate_concern:party_closeGreen Party                          0.398
33 issueeast_germans:party_closeGreen Party                            -0.128
34 issueecon_inequality:party_closeGreen Party                          0.087
35 issuegay_marriage:party_closeGreen Party                             0.244
36 issuerights_indep_integration:party_closeGreen Party                 0.071
37 issueclimate_concern:party_closeLeft Party                           0.372
38 issueeast_germans:party_closeLeft Party                             -0.03
39 issueecon_inequality:party_closeLeft Party                           0.2
40 issuegay_marriage:party_closeLeft Party                              0.29
41 issuerights_indep_integration:party_closeLeft Party                  0.14
42 issueclimate_concern:party_closeNo party                             0.204
43 issueeast_germans:party_closeNo party                               -0.042
44 issueecon_inequality:party_closeNo party                             0.032
45 issuegay_marriage:party_closeNo party                                0.117
46 issuerights_indep_integration:party_closeNo party                    0.065
47 issueclimate_concern:party_closeOther party                          0.182
48 issueeast_germans:party_closeOther party                            -0.038
49 issueecon_inequality:party_closeOther party                          0.107
50 issuegay_marriage:party_closeOther party                             0.148
51 issuerights_indep_integration:party_closeOther party                 0.052
52 issueclimate_concern:party_closeRefuse to say/No answer              0.185
53 issueeast_germans:party_closeRefuse to say/No answer                -0.036
54 issueecon_inequality:party_closeRefuse to say/No answer             -0.002
55 issuegay_marriage:party_closeRefuse to say/No answer                 0.151
56 issuerights_indep_integration:party_closeRefuse to say/No answer     0.098
57 issueclimate_concern:party_closeSPD                                  0.227
58 issueeast_germans:party_closeSPD                                    -0.14
59 issueecon_inequality:party_closeSPD                                  0.033
60 issuegay_marriage:party_closeSPD                                     0.092
61 issuerights_indep_integration:party_closeSPD                        -0.057
   std.error p.value
```

```
      <dbl>   <dbl>
 1    0.015   0
 2    0.018   0
 3    0.018   0.637
 4    0.018   0.009
 5    0.018   0
 6    0.018   0.454
 7    0.036   0.775
 8    0.02    0.016
 9    0.036   0.621
10    0.023   0.353
11    0.027   0.272
12    0.021   0.019
13    0.044   0.148
14    0.062   0.585
15    0.022   0
16    0.005   0.787
17    0.043   0.002
18    0.043   0.931
19    0.043   0.016
20    0.043   0.086
21    0.043   0.162
22    0.024   0
23    0.024   0
24    0.024   0.617
25    0.024   0
26    0.024   0.837
27    0.044   0
28    0.044   0.409
29    0.044   0.341
30    0.044   0.016
31    0.044   0.159
32    0.027   0
33    0.027   0
34    0.027   0.001
35    0.027   0
36    0.027   0.009
37    0.032   0
38    0.032   0.351
39    0.032   0
40    0.032   0
41    0.032   0
42    0.026   0
43    0.026   0.108
44    0.026   0.215
45    0.026   0
46    0.026   0.012
47    0.054   0.001
48    0.054   0.48
49    0.054   0.047
50    0.054   0.006
51    0.054   0.338
52    0.078   0.018
53    0.078   0.641
54    0.078   0.985
55    0.078   0.054
56    0.078   0.213
57    0.026   0
58    0.026   0
59    0.026   0.202
60    0.026   0
61    0.026   0.03

R²  Null: marginal = 0.027, conditional = 0.271  |  Full: marginal = 0.096, conditional = 0.308
```

```
=== H6 Weights (issue weights: w): party_close × issue interaction ===
  chi^2(54) = 805.895,  p = 0.0000  ***
```

And again, per issue:

```r
run_H_per_issue("w",  "party_close", "H6 Weights", use_lmer = TRUE)
```

```
-- H6 Weights per issue (issue weight: w) --
climate_concern  R^2=0.179/0.654  LRT chi^2(9)=292.84  p=0.0000 ***
gay_marriage  R^2=0.087/0.601  LRT chi^2(9)=143.48  p=0.0000 ***
rights_indep_integration  R^2=0.018/0.428  LRT chi^2(9)=29.60  p=0.0005 ***
econ_inequality  R^2=0.066/0.572  LRT chi^2(9)=111.69  p=0.0000 ***
regulate_internet  R^2=0.032/0.488  LRT chi^2(9)=55.12  p=0.0000 ***
east_germans  R^2=0.011/0.656  LRT chi^2(9)=17.96  p=0.0356 *
```

```r
run_H_joint    ("w", "lr", "H7 Weights", use_lmer = TRUE)
```

```
 Model    AIC    BIC  logLik npar delta_AIC delta_BIC
  Null 2457.1 2524.2 -1219.6    7     392.7     347.9
  Full 2064.4 2176.3 -1017.2   13       0.0       0.0
# A tibble: 13 x 4
   term                             estimate std.error p.value
   <chr>                               <dbl>     <dbl>   <dbl>
 1 (Intercept)                         0.542     0.016   0
 2 issueclimate_concern                0.275     0.019   0
 3 issueeast_germans                  -0.086     0.019   0
 4 issueecon_inequality                0.216     0.019   0
 5 issuegay_marriage                   0.225     0.019   0
 6 issuerights_indep_integration       0.088     0.019   0
 7 lr                                  0.016     0.031   0.597
 8 wave2                               0.001     0.005   0.905
 9 issueclimate_concern:lr            -0.439     0.036   0
10 issueeast_germans:lr                0.05      0.036   0.16
11 issueecon_inequality:lr            -0.265     0.036   0
12 issuegay_marriage:lr               -0.399     0.036   0
13 issuerights_indep_integration:lr   -0.097     0.036   0.007

R²  Null: marginal = 0.027, conditional = 0.271  |  Full: marginal = 0.062, conditional = 0.289
=== H7 Weights (issue weights: w): lr × issue interaction ===
  chi^2(6) = 404.659,  p = 0.0000  ***
```

And again, per issue:

```r
run_H_per_issue("w",  "lr", "H7 Weights", use_lmer = TRUE)
```

```
-- H7 Weights per issue (issue weight: w) --
climate_concern  R^2=0.075/0.670  LRT chi^2(1)=119.39  p=0.0000 ***
gay_marriage  R^2=0.070/0.604  LRT chi^2(1)=114.72  p=0.0000 ***
rights_indep_integration  R^2=0.005/0.427  LRT chi^2(1)=8.10  p=0.0044 **
econ_inequality  R^2=0.042/0.573  LRT chi^2(1)=68.19  p=0.0000 ***
regulate_internet  R^2=0.001/0.483  LRT chi^2(1)=0.04  p=0.8321 ns
east_germans  R^2=0.001/0.655  LRT chi^2(1)=0.96  p=0.3260 ns
```

```r
run_H_joint    ("w", "age", "H8 Weights", use_lmer = TRUE)
```

```
 Model    AIC    BIC  logLik npar delta_AIC delta_BIC
  Null 2450.0 2517.0 -1216.0    7      60.6      15.9
  Full 2389.4 2501.1 -1179.7   13       0.0       0.0
# A tibble: 13 x 4
   term                               estimate std.error p.value
   <chr>                                 <dbl>     <dbl>   <dbl>
 1 (Intercept)                           0.439     0.023   0
 2 issueclimate_concern                  0.104     0.026   0
 3 issueeast_germans                    -0.084     0.026   0.001
 4 issueecon_inequality                  0.091     0.026   0
```

```
 5 issuegay_marriage                    0.129     0.026   0
 6 issuerights_indep_integration        0.154     0.026   0
 7 age                                  0.002     0       0
 8 wave2                                0         0.005   0.939
 9 issueclimate_concern:age            -0.001     0.001   0.078
10 issueeast_germans:age                0         0.001   0.355
11 issueecon_inequality:age             0         0.001   0.866
12 issuegay_marriage:age               -0.002     0.001   0
13 issuerights_indep_integration:age   -0.002     0.001   0

R²  Null: marginal = 0.026, conditional = 0.270  |  Full: marginal = 0.035, conditional = 0.273
=== H8 Weights (issue weights: w): age × issue interaction ===
  chi^2(6) = 72.622,  p = 0.0000  ***
```

And again, per issue:

```r
run_H_per_issue("w",  "age", "H8 Weights", use_lmer = TRUE)
```

```
-- H8 Weights per issue (issue weight: w) --
climate_concern  R^2=0.005/0.682  LRT chi^2(1)=6.93  p=0.0085 **
gay_marriage  R^2=0.000/0.606  LRT chi^2(1)=0.25  p=0.6190 ns
rights_indep_integration  R^2=0.000/0.427  LRT chi^2(1)=0.01  p=0.9076 ns
econ_inequality  R^2=0.016/0.577  LRT chi^2(1)=23.59  p=0.0000 ***
regulate_internet  R^2=0.017/0.484  LRT chi^2(1)=24.81  p=0.0000 ***
east_germans  R^2=0.020/0.657  LRT chi^2(1)=27.84  p=0.0000 ***
```

3.3 Aim 3

Aim 3: Assess the role of social circle belief variation

For Aim 3, we test whether including social circle belief variation improves predictions for participants' issue weights (inferred from the map distances), their reported issue importance, likeability of voters, and polarisation perceptions. In addition, we explore how social circle belief improves predictions of the map distances between individuals.

3.3.1 H9

3.3.1.1 H9. Social circle variance predicts issue weights.

We model issue weights of participant n as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot \mathrm{var}_{n,q} + u_q + \epsilon_{n,q} \,. \tag{14}$$

The prediction is $\beta_q < 0$. Issues are assigned higher weight when associated with lower social circle variance.

CHANGE TO PREREG: I am using standard deviation instead of variance now!!!

$$\alpha_{n,q} = 1/6 + \beta_q \cdot \mathrm{var}_{n,q} + u_q + \epsilon_{n,q} \,. \tag{15}$$

```r
run_h9 <- function(k = "exp_alpha_fitAllDots", variance_type = "std") {
  var_col <- paste0(variance_type, "_socialCircle_ops")
  alpha_cols <- paste0(k, "_", questions_sc)

  df_h9 <- df_p |>
    filter(wave %in% c(1, 2)) |>
    select(id, wave, all_of(paste0(var_col, "_", questions_sc)), all_of(alpha_cols)) |>
    pivot_longer(
      cols          = matches(paste0("^(", var_col, "|", k ,")_")),
      names_to      = c(".value", "issue"),
      names_pattern = paste0("^(", var_col, "|", k, ")_(.*)")
    ) |>
    rename(var_sc = all_of(var_col), alpha = all_of(k)) |>
    filter(!is.na(alpha), !is.na(var_sc))

  m  <- lmer(alpha ~ var_sc + wave + (1 | issue), data = df_h9)
  cf <- lmer_coef(m, "var_sc")
  print_one_sided(
```

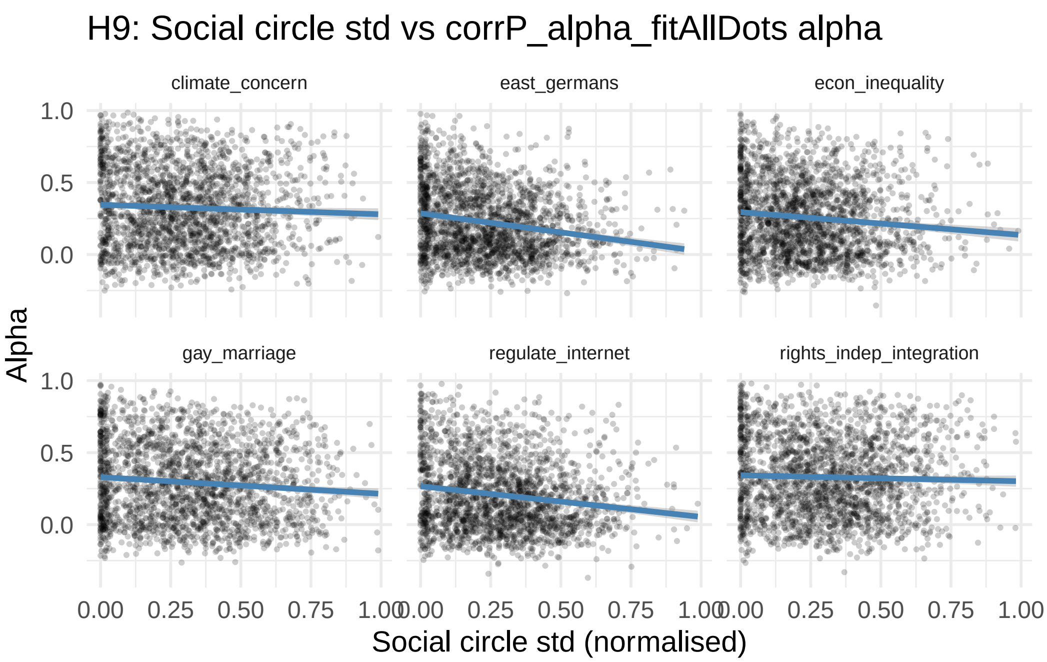
```r
    sprintf("H9: Social circle %s predicts issue weights (%s)", variance_type, k),
    cf$beta, cf$se, cf$t, direction = "negative"
  )

  # Per-issue
  cat("Per-issue breakdown:\n")
  purrr::walk(questions_sc, function(q) {
    alpha_col <- paste0(k, "_", q)
    df_q <- df_p |>
      filter(wave %in% c(1, 2)) |>
      select(id, wave,
             var_sc = all_of(paste0(var_col, "_", q)),
             alpha  = all_of(alpha_col)) |>
      filter(!is.na(alpha), !is.na(var_sc))
    m_q  <- lmer(alpha ~ var_sc + wave + (1 | id), data = df_q)
    cf_q <- lmer_coef(m_q, "var_sc")
    cat(sprintf("  %-35s  beta=%7.4f  t=%6.3f  p=%.4f  %s\n",
                q, cf_q$beta, cf_q$t, cf_q$p2, sig_stars(cf_q$p2)))
  })
  cat("\n")
}

run_h9(paste0("exp_alpha_",fitmode) ,   "var")
run_h9(paste0("corrP_alpha_",fitmode), "std")
```

```
=== H9: Social circle var predicts issue weights (exp_alpha_fitAllDots) ===
  Estimate  =  -0.0786
  SE        =   0.0137
  t         =  -5.7491
  df        = infty
  p (one-sided, beta < 0) = 0.0000
  Supported : YES

Per-issue breakdown:
  climate_concern                      beta=-0.0390  t=-1.134  p=0.2568  ns
  gay_marriage                         beta=-0.1000  t=-3.546  p=0.0004  ***
  rights_indep_integration             beta= 0.0180  t= 0.536  p=0.5918  ns
  econ_inequality                      beta=-0.0780  t=-1.958  p=0.0504  .
  regulate_internet                    beta=-0.1374  t=-4.260  p=0.0000  ***
  east_germans                         beta=-0.2179  t=-5.117  p=0.0000  ***

=== H9: Social circle std predicts issue weights (corrP_alpha_fitAllDots) ===
  Estimate  =  -0.1352
  SE        =   0.0116
  t         = -11.7050
  df        = infty
  p (one-sided, beta < 0) = 0.0000
  Supported : YES

Per-issue breakdown:
  climate_concern                      beta=-0.0997  t=-3.373  p=0.0008  ***
  gay_marriage                         beta=-0.1451  t=-5.472  p=0.0000  ***
  rights_indep_integration             beta=-0.0746  t=-2.633  p=0.0085  **
  econ_inequality                      beta=-0.1673  t=-5.342  p=0.0000  ***
  regulate_internet                    beta=-0.2196  t=-7.577  p=0.0000  ***
  east_germans                         beta=-0.2537  t=-8.779  p=0.0000  ***
```

```r
k <- paste0("corrP_alpha_", fitmode)
variance_type <- "std"
alpha_cols <- paste0(k, "_", questions_sc)

df_h9_plot <- df_p |>
  filter(wave %in% c(1, 2)) |>
  pivot_longer(
```

```
    cols          = matches(paste0("^(std_socialCircle_ops|", k, ")_")),
    names_to      = c(".value", "issue"),
    names_pattern = paste0("^(std_socialCircle_ops|", k, ")_(.*)")
  ) |>
  rename(var_sc = all_of(paste0(variance_type, "_socialCircle_ops")), alpha = all_of(k)) |>
  filter(!is.na(alpha), !is.na(var_sc))

ggplot(df_h9_plot, aes(x = var_sc, y = alpha)) +
  geom_point(alpha = 0.2, size = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "steelblue") +
  facet_wrap(~ issue, ncol = 3) +
  labs(title = sprintf("H9: Social circle %s vs %s alpha", variance_type, k),
       x = sprintf("Social circle %s (normalised)", variance_type), y = "Alpha") +
  theme_minimal() +
  theme(strip.text = element_text(size = 7))
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
ggsave(sprintf("figs/h9_plot_%s_%s.png", k, variance_type), width = 16/2.54, height = 9/2.54)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

3.3.2 H10

3.3.2.1 H10. Social circle variance predicts reported issue importance.

We model reported issue importance of participant n as

$$w_{n,q} = \beta_0 + \beta_q \cdot \mathrm{var}_{n,q} + u_q + v_n + \epsilon_{n,q} \ . \tag{16}$$

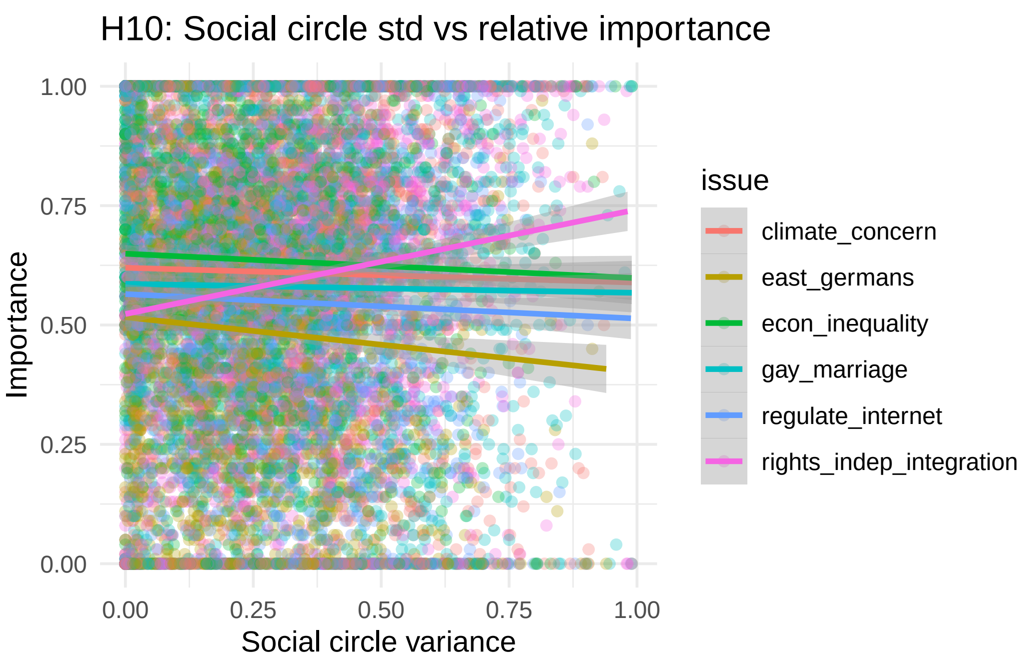
where v_n is the average importance of all issues assigned by participant n.

The prediction is $\beta_q < 0$. Issues are rated as more important when social circle variance is smaller.

```
run_h10 <- function(variance_type = "std") {
  var_col       <- paste0(variance_type, "_socialCircle_ops")

  df_h10 <- df_p |>
    filter(wave %in% c(1, 2)) |>
    pivot_longer(
      cols          = c(all_of(paste0("w_", questions_sc)),
                        all_of(paste0(var_col, "_", questions_sc))),
      names_to      = c(".value", "issue"),
      names_pattern = paste0("^(", var_col, "|w)_(.*)")
    ) |>
```

```r
    rename(var_sc = all_of(var_col))

  m_h10 <- lmer(w ~ var_sc + wave + (1 | issue) + (1 | id), data = df_h10)
  cf_h10 <- lmer_coef(m_h10, "var_sc")

  print_one_sided(
    sprintf("H10: Social circle %s predicts reported issue importance", variance_type),
    cf_h10$beta, cf_h10$se, cf_h10$t, direction = "negative"
  )

  # Per-issue
  cat("Per-issue breakdown:\n")
  purrr::walk(questions_sc, function(q) {
    df_q <- df_h10 |> filter(issue == q) |> select(id, wave, issue, w, var_sc)
    m_q  <- lmer(w ~ var_sc + wave + (1 | id), data = df_q)
    cf_q <- lmer_coef(m_q, "var_sc")
    cat(sprintf("  %-35s  beta=%7.4f  t=%6.3f  p=%.4f  %s\n",
                q, cf_q$beta, cf_q$t, cf_q$p2, sig_stars(cf_q$p2)))
  })

  ggplot(df_h10, aes(x = var_sc, y = w, color = issue)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "Social circle variance", y = "Importance",
      title = sprintf("H10: Social circle %s vs relative importance", variance_type)) +
  theme_minimal()
}

run_h10(variance_type="std")
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
=== H10: Social circle std predicts reported issue importance ===
  Estimate  =   0.0053
  SE        =   0.0141
  t         =   0.3748
  df        = infty
  p (one-sided, beta < 0) = 0.6461
  Supported : NO

Per-issue breakdown:
  climate_concern                      beta=-0.0079  t=-0.260  p=0.7952  ns
  gay_marriage                         beta=-0.0176  t=-0.589  p=0.5559  ns
```

```
  rights_indep_integration              beta= 0.1817  t= 5.917  p=0.0000  ***
  econ_inequality                       beta=-0.0584  t=-1.919  p=0.0551  .
  regulate_internet                     beta=-0.0528  t=-1.704  p=0.0885  .
  east_germans                          beta=-0.1060  t=-3.146  p=0.0017  **
```

3.3.3 H11

3.3.3.1 H11. Overall social circle variance helps predict likeability rating of typical voters.

In H2 (equation 2), we modelled the participant n's likeability rating of the typical voter of political party r as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \tag{17}$$

Here, we extend this model as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} + v_n + \epsilon_{n,r} \tag{18}$$

where the new β_2-term represents the influence of overall social circle belief variation on likeability.

The prediction is $\beta_2 > 0$ and that equation 19 improves the predictions of the baseline model from **H2** (equation 2/17) using R2, AIC, and BIC as model quality indicators. Voters are more liked in general if the participant's overall social circle variance is higher.

TODO: This hypothesis is kind of wrong. We have 1 avgsocialcirclestd per person and 7 pixeldist and sympathy for 7 parties. But the avgsocialcirclestd is already included in the random intercept.

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} REMOVED(+v_n) + \epsilon_{n,r} \tag{19}$$

```r
variance_type <- "std"

avg_var_sc <- df_p |>
  filter(wave == 2) |>
  select(id, all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc))) |>
  drop_na() |>
  mutate(avg_var_sc = rowMeans(across(-id))) |>
  select(id, avg_var_sc)

df_h11 <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, sympathy, party, pixel_dist) |>
  left_join(avg_var_sc, by = "id") |>
  mutate(
    # pixel_dist_z = as.numeric(scale(pixel_dist)),
         avg_var_sc_z = as.numeric(scale(avg_var_sc)))

m_base_h11 <- lmer(sympathy ~ pixel_dist + (1 | id),
                   data = df_h11, REML = FALSE)
m_ext_h11  <- lmer(sympathy ~ pixel_dist + avg_var_sc_z + (1 | id),
                   data = df_h11, REML = FALSE)

cf_h11 <- lmer_coef(m_ext_h11, "avg_var_sc_z")

# One-sided test: H1 = avg_var_sc_z has a positive effect
p_one_h11 <- if (cf_h11$beta > 0) {
  cf_h11$p2 / 2
} else {
  1 - cf_h11$p2 / 2
}

r2_base <- r2(m_base_h11)
r2_ext  <- r2(m_ext_h11)

cat("=== H11: Social circle variance predicts likeability (beyond map distance) ===\n")
cat(sprintf(" beta (pixel_dist) = %.4f\n",
            fixef(m_ext_h11)["pixel_dist"]))
cat(sprintf(" beta (avg_var_sc_z) = %.4f\n",
```

```
            cf_h11$beta))
cat(sprintf(" SE = %.4f\n",
            cf_h11$se))
cat(sprintf(" t = %.4f\n",
            cf_h11$t))
cat(sprintf(" df (Satterthwaite) = %.2f\n",
            cf_h11$df))
cat(sprintf(" p (two-sided) = %.6f\n",
            cf_h11$p2))
cat(sprintf(" p (one-sided, beta > 0) = %.6f\n",
            p_one_h11))
cat(sprintf(" Supported: %s\n",
            ifelse(p_one_h11 < 0.05 && cf_h11$beta > 0,
                   "YES", "NO")))
cat(sprintf(" R² marginal base/ext = %.4f / %.4f (Delta=%+.4f)\n",
            r2_base$R2_marginal,
            r2_ext$R2_marginal,
            r2_ext$R2_marginal - r2_base$R2_marginal))
cat(sprintf(" R² conditional base/ext = %.4f / %.4f (Delta=%+.4f)\n",
            r2_base$R2_conditional,
            r2_ext$R2_conditional,
            r2_ext$R2_conditional - r2_base$R2_conditional))
cat(sprintf(" AIC base/ext = %.2f / %.2f (Delta=%+.2f)\n",
            AIC(m_base_h11),
            AIC(m_ext_h11),
            AIC(m_ext_h11) - AIC(m_base_h11)))
cat(sprintf(" BIC base/ext = %.2f / %.2f (Delta=%+.2f)\n\n",
            BIC(m_base_h11),
            BIC(m_ext_h11),
            BIC(m_ext_h11) - BIC(m_base_h11)))
```

```
=== H11: Social circle variance predicts likeability (beyond map distance) ===
 beta (pixel_dist) = -1.1115
 beta (avg_var_sc_z) = -0.0135
 SE = 0.0052
 t = -2.5813
 df (Satterthwaite) = 854.38
 p (two-sided) = 0.010010
 p (one-sided, beta > 0) = 0.994995
 Supported: NO
 R² marginal base/ext = 0.4208 / 0.4211 (Delta=+0.0002)
 R² conditional base/ext = 0.5802 / 0.5775 (Delta=-0.0027)
 AIC base/ext = 313.82 / 303.83 (Delta=-9.99)
 BIC base/ext = 341.08 / 337.89 (Delta=-3.19)
```

3.3.4 H12a

3.3.4.1 H12a. Social circle variance predicts perceived political polarisation on a given issue.

We model perceived polarisation on issue q as

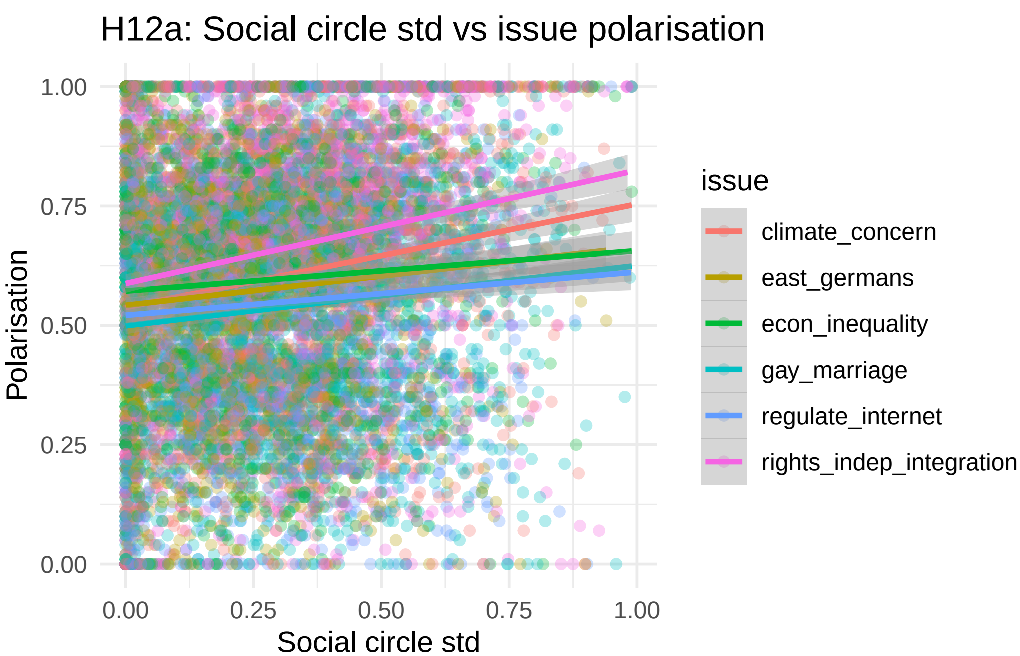
$$P_{n,q} = \beta_0 + \beta_q \cdot var_{n,q} + u_n + \epsilon_{n,q}. \tag{20}$$

The prediction is $\beta_q < 0$. Participants with less variation in their social circles with respect to issue q perceive higher polarisation on that issue. %We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```
variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")

df_h12a <- df_p |>
  filter(wave %in% c(1, 2)) |>
  select(-P_tot) |>
  pivot_longer(
    cols          = c(all_of(paste0("P_", questions_sc)),
                      all_of(paste0(var_col, "_", questions_sc))),
```

```
    names_to      = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, "|P)_(.*)")
  ) |>
  rename(var_sc = all_of(var_col))

m_h12a <- lmer(P ~ var_sc + wave + (1 | issue) + (1 | id), data = df_h12a)
cf_h12a <- lmer_coef(m_h12a, "var_sc")

print_one_sided(
  sprintf("H12a: Social circle %s predicts perceived polarisation (per issue)", variance_type),
  cf_h12a$beta, cf_h12a$se, cf_h12a$t, direction = "negative"
)

ggplot(df_h12a, aes(x = var_sc, y = P, color = issue)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = sprintf("Social circle %s", variance_type), y = "Polarisation",
       title = sprintf("H12a: Social circle %s vs issue polarisation", variance_type)) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
=== H12a: Social circle std predicts perceived polarisation (per issue) ===
  Estimate   =   0.1685
  SE         =   0.0117
  t          =  14.4028
  df         = infty
  p (one-sided, beta < 0) = 1.0000
  Supported : NO
```

3.3.5 H12b

3.3.5.1 H12b. Social circle variance predicts perceived political polarisation.

We model overall perceived polarisation as

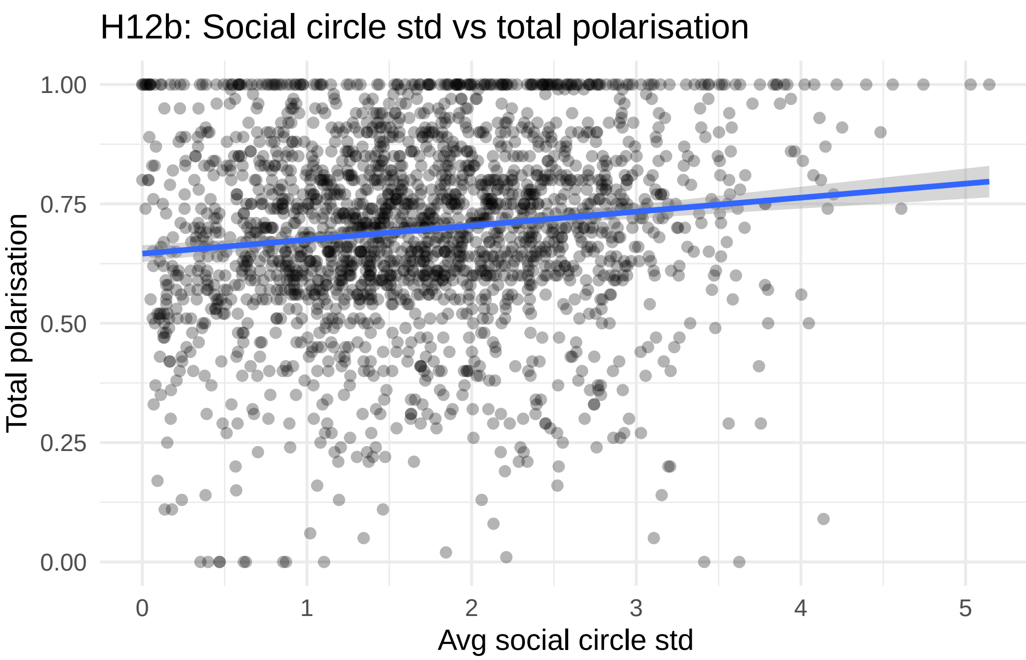
$$P_{n,tot} = \beta_0 + \beta_1 \cdot \sum_q var_{n,q} + \epsilon_n. \tag{21}$$

The prediction is \$ _1 < 0\$. Participants with less variation in their social circles averaged over all issues perceive more political polarisation. %We test this via a t-test with significance threshold $\alpha = 0.05$.

```
df_h12b <- df_p |>
  select(id, wave, all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc)), P_tot) |>
  drop_na() |>
  mutate(avg_var_sc = rowSums(across(all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc)))))
```

```
m_h12b   <- lmer(P_tot ~ avg_var_sc + (1 | id), data = df_h12b, REML = TRUE)
n_p_h12b <- n_distinct(df_h12b$id)
cf_h12b  <- lmer_coef(m_h12b, "avg_var_sc")
t_h12b   <- cf_h12b$beta / cf_h12b$se

print_one_sided(
  "H12b: Overall social circle variance predicts total perceived polarisation",
  cf_h12b$beta, cf_h12b$se, t_h12b, df = n_p_h12b - 2,
  direction = "negative"
)

ggplot(df_h12b, aes(x = avg_var_sc, y = P_tot)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = sprintf("Avg social circle %s", variance_type), y = "Total polarisation",
       title = sprintf("H12b: Social circle %s vs total polarisation", variance_type)) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
=== H12b: Overall social circle variance predicts total perceived polarisation ===
  Estimate  =   0.0234
  SE        =   0.0050
  t         =   4.6493
  df        = 1225.0
  p (one-sided, beta < 0) = 1.0000
  Supported : NO
```

3.3.6 H13

3.3.6.1 H13. Reported issue importance predicts perceived distance.

In equation **??**, we fix the issue weights to $\alpha_{n,q} = w_{n,q}$. Then, we model the perceived distance between two individuals i and j (using the exponential kernel; see equations **??** and **??**) as

$$d_{n,ij} = 1 - \exp\left(-\hat{b}_n \cdot \left(\sum_q w_{n,q} \cdot |x_{n,i,q} - x_{n,j,q}|\right)^{\hat{c}_n}\right) \tag{22}$$

with fit parameters $\hat{b}_n$, and $\hat{c}_n$.

The prediction is that this model (with $\hat{\alpha}_{n,q} = w_{n,q}$) outperforms (i) an unweighted model with $\hat{\alpha}_{n,q} = 1/6$ and (ii) a population weight model with $\hat{\alpha}_{n,q} = 1/n \cdot \sum_n w_{n,q}$. We use R2, AIC, and BIC as model quality indicators.

3.3.6.2 Step 1: Make predictions

```
linear_kernel <- function(x, a, b) a + b * x
exp_kernel    <- function(x, a, b) 1 - exp(-a * (x ^ b))
log_kernel    <- function(x, a, b) 1 / (1 + exp(-b * (x - a)))

r2_vec        <- function(y, pm) {
  ss_tot <- sum((y - mean(y, na.rm = TRUE))^2, na.rm = TRUE)
  1 - colSums((pm - y)^2, na.rm = TRUE) / ss_tot
}
compute_aic_bic <- function(mse, k, n) {
  rss <- mse * n
  tibble(aic = n * log(rss / n) + 2 * k, bic = n * log(rss / n) + k * log(n))
}

col_alpha_corrP <- paste0("corrP_alpha_", fitmode, "_",    questions_sc)
col_alpha_corrS <- paste0("corrS_alpha_", fitmode, "_",    questions_sc)
col_alpha_exp <- paste0("exp_alpha_", fitmode, "_",    questions_sc)
col_alpha_lin <- paste0("linear_alpha_", fitmode, "_", questions_sc)
col_alpha_log <- paste0("logistic_alpha_", fitmode, "_", questions_sc)

alphas_pop_lin <- df_p |> select(all_of(col_alpha_lin)) |> colMeans()
alphas_pop_exp <- df_p |> select(all_of(col_alpha_exp)) |> colMeans()
alphas_pop_log <- df_p |> select(all_of(col_alpha_log)) |> colMeans()

df_h13_base <- df_diff |>
  filter(wave %in% c(1, 2)) |>
  left_join(df_p |> select(id, wave, all_of(col_weights),
                           all_of(col_alpha_lin), all_of(col_alpha_log), all_of(col_alpha_exp), all_of(col_alph
            by = c("id", "wave"), suffix = c("", ".drop")) |>
  select(-ends_with(".drop"))

.d         <- as.matrix(df_h13_base[, col_deltas])
.a_uniform <- matrix(1 / length(questions_sc), nrow = nrow(df_h13_base), ncol = length(questions_sc))
.a_pop_exp <- matrix(alphas_pop_exp, nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_pop_lin <- matrix(alphas_pop_lin, nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_pop_log <- matrix(alphas_pop_log, nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_corrP <- as.matrix(df_h13_base[, col_alpha_corrP])
.a_corrP_rel <- {
  cP <- as.matrix(df_h13_base[, col_alpha_corrP])
  cP_sum <- rowSums(abs(cP))
  ifelse(cP_sum == 0, 0, cP / cP_sum)
}
.a_fit_exp <- as.matrix(df_h13_base[, col_alpha_exp])
.a_fit_lin <- as.matrix(df_h13_base[, col_alpha_lin])
.a_fit_log <- as.matrix(df_h13_base[, col_alpha_log])
.a_weights <- {
  w <- as.matrix(df_h13_base[, col_weights])
  w_sum <- rowSums(w, na.rm = TRUE)
  ifelse(w_sum == 0, 0, w / w_sum)
}

pred_cols <- c("weights", "population_exp",
               "population_linear", "population_logistic", "uniform_exp",
               "uniform_linear", "fit_linear",
               "fit_exp", "corrP", "corrPrel")
p <- function(prefix, n) {
  df_h13_base[[paste0(prefix, "_", fitmode, "_param", n)]]
}

df_h13 <- df_h13_base |>
  mutate(
    uniform_exp          = exp_kernel(rowSums(.d * .a_uniform),    p("exp", 1),      p("exp", 2)),
    uniform_linear        = linear_kernel(rowSums(.d * .a_uniform),   p("linear", 1),    p("linear", 2)),
```

```r
    uniform_logistic      = log_kernel(rowSums(.d * .a_uniform),   p("logistic", 1),  p("logistic", 2)),
    population_exp        = exp_kernel(rowSums(.d * .a_pop_exp),      p("exp", 1),      p("exp", 2)),
    population_linear     = linear_kernel(rowSums(.d * .a_pop_lin),   p("linear", 1),    p("linear", 2)),
    population_logistic   = log_kernel(rowSums(.d * .a_pop_log),   p("logistic", 1),  p("logistic", 2)),
    fit_exp                = exp_kernel(rowSums(.d * .a_fit_exp),      p("exp", 1),      p("exp", 2)),
    fit_linear              = linear_kernel(rowSums(.d * .a_fit_lin),   p("linear", 1),    p("linear", 2)),
    fit_logistic            = log_kernel(rowSums(.d * .a_fit_log),   p("logistic", 1),  p("logistic", 2)),
    weights                = exp_kernel(rowSums(.d * .a_weights),      p("exp", 1),      p("exp", 2)),
    corrP                  = exp_kernel(rowSums(.d * .a_corrP),        p("exp", 1),      p("exp", 2)),
    corrPrel               = exp_kernel(rowSums(.d * .a_corrP_rel),    p("exp", 1),      p("exp", 2))
  ) |>
  mutate(across(all_of(pred_cols),
              ~ pixel_dist - .x,
              .names = "err_{.col}"))
```

3.3.6.3 Step 2: Evaluate predictions

```r
y        <- df_h13$pixel_dist
n_obs    <- sum(!is.na(y))
n_pw     <- df_h13 |> distinct(id, wave) |> nrow()
n_q      <- length(questions_sc)
pred_mat <- df_h13 |> select(all_of(pred_cols)) |> as.matrix()
colnames(pred_mat) <- str_remove(pred_cols, "predicted_d_")

k_lookup <- tribble(
  ~model,            ~k,
  "weights",         0,
  "population_exp",  n_q,
  "population_lin",  n_q,
  "population_logistic",  n_q,
  "uniform_exp",     0,
  "uniform_lin",     0,
  "fit_exp",         (n_q) * n_pw,
  "fit_linear",      (n_q) * n_pw,
  "fit_logistic",      (n_q) * n_pw,
  "corrP",            0,
  "corrPrel",         0
)
baseline_sq <- (pred_mat[, "uniform_exp"] - y)^2

comparison_tbl <- tibble(
  model = colnames(pred_mat),
  mse   = colMeans((pred_mat - y)^2, na.rm = TRUE),
  rmse  = sqrt(mse),
  r2    = r2_vec(y, pred_mat)
) |>
  mutate(
    rel_improve          = (mse[model == "uniform_exp"] - mse) / mse[model == "uniform_exp"],
    p_wilcox             = map_dbl(model, function(m) {
      if (m == "uniform_exp") return(NA_real_)
      wilcox.test((pred_mat[, m] - y)^2, baseline_sq, paired = TRUE)$p.value
    }),
    mean_diff_vs_uniform = map_dbl(model, function(m) {
      if (m == "uniform_exp") return(NA_real_)
      mean((pred_mat[, m] - y)^2 - baseline_sq, na.rm = TRUE)
    })
  ) |>
  left_join(k_lookup, by = "model") |>
  mutate(
    aic_bic = map2(mse, k, ~ compute_aic_bic(.x, .y, n_obs)),
    aic     = map_dbl(aic_bic, "aic"),
    bic     = map_dbl(aic_bic, "bic")
  ) |>
```

```
  select(-aic_bic, -k) |>
  arrange(mse)

print(comparison_tbl)
```

```
# A tibble: 10 x 9
   model                  mse  rmse      r2 rel_improve  p_wilcox
   <chr>                <dbl> <dbl>   <dbl>       <dbl>     <dbl>
 1 fit_linear          0.0162 0.127  0.509      0.235    0
 2 fit_exp             0.0164 0.128  0.503      0.226    0
 3 population_exp      0.0212 0.146  0.360      0.00251  3.00e-74
 4 uniform_exp         0.0212 0.146  0.358      0       NA
 5 population_logistic 0.0233 0.153  0.296     -0.0964   0
 6 uniform_linear      0.0250 0.158  0.244     -0.178    2.52e-13
 7 population_linear   0.0252 0.159  0.240     -0.185    4.22e- 2
 8 corrPrel            0.0268 0.164  0.297     -0.262    0
 9 weights             0.0324 0.180  0.0197    -0.527    0
10 corrP               0.0493 0.222 -0.357     -1.32     0
   mean_diff_vs_uniform       aic       bic
                  <dbl>     <dbl>     <dbl>
 1           -0.00499   -1111103.  -976494.
 2           -0.00480   -1107880.  -973271.
 3           -0.0000532 -1063411. -1063347.
 4           NA         -1062730. -1062730.
 5            0.00205   -1037335. -1037272.
 6            0.00377         NA        NA
 7            0.00392         NA        NA
 8            0.00608    -998419.  -998419.
 9            0.0112     -945927.  -945927.
10            0.0282     -830593.  -830593.
```

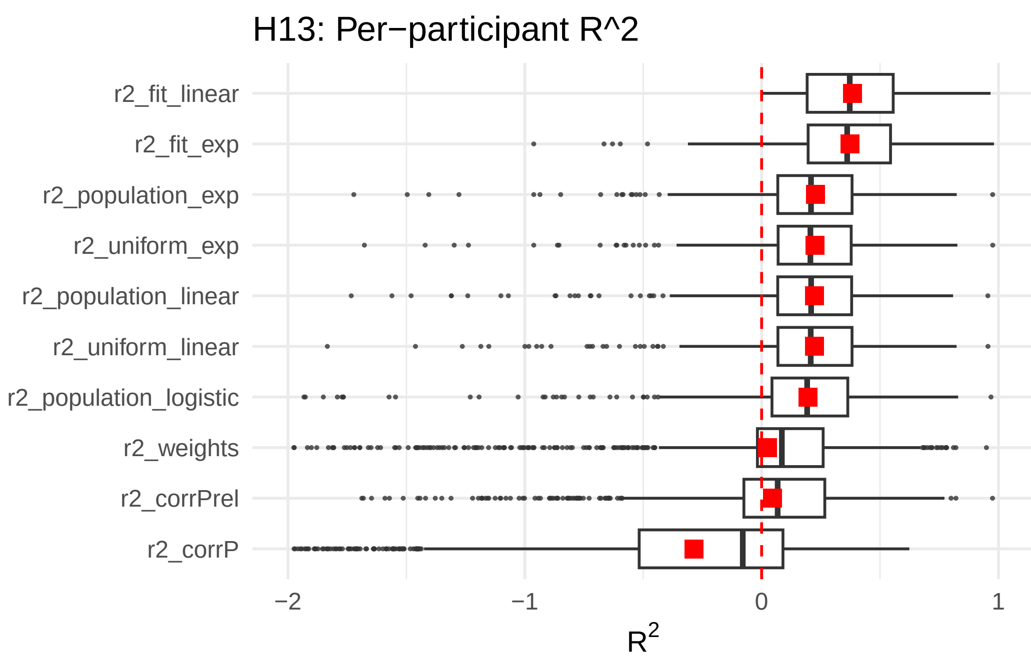
3.3.6.4 Step 3: Analysis of predictions

```
err_cols <- paste0("err_", pred_cols)

# Per-participant R^2
per_person_r2 <- df_h13 |>
  group_by(id) |>
  summarise(across(all_of(pred_cols), function(x) {
    v <- !is.na(x) & !is.na(pixel_dist)
    if (sum(v) < 3) return(NA_real_)
    ss_res <- sum((pixel_dist[v] - x[v])^2)
    ss_tot <- sum((pixel_dist[v] - mean(pixel_dist[v]))^2)
    if (ss_tot == 0) NA_real_ else 1 - ss_res / ss_tot
  }, .names = "r2_{.col}")) |>
  pivot_longer(starts_with("r2_"), names_to = "model", values_to = "r2") |>
  mutate(model = str_remove(model, "r2_predicted_d_"))

p1 <- ggplot(per_person_r2, aes(x = reorder(model, r2, median, na.rm = TRUE), y = r2)) +
  geom_boxplot(outlier.alpha = 0.8, outlier.size = 0.2) +
  stat_summary(fun = mean, geom = "point", shape = 15, size = 3, colour = "red") +
  geom_hline(yintercept = 0, colour = "red", linetype = "dashed") +
  coord_flip() + ylim(-2, 1) +
  labs(title = "H13: Per-participant R^2", x = NULL, y = expression(R^2)) +
  theme_minimal()

# Predicted vs observed
p2 <- df_h13 |>
  slice_sample(n = 10000) |>
  select(pixel_dist, all_of(pred_cols)) |>
  pivot_longer(-pixel_dist, names_to = "model", values_to = "predicted") |>
  mutate(model = str_remove(model, "predicted_d_")) |>
  ggplot(aes(pixel_dist, predicted)) +
  geom_point(alpha = 0.05, size = 0.3) +
```

```
  geom_abline(slope = 1, intercept = 0, colour = "red") +
  geom_smooth(method = "gam", formula = y ~ s(x, bs = "cs"), se = FALSE, colour = "steelblue") +
  facet_wrap(~ model, ncol = 4) + ylim(0, 1) + coord_equal() +
  labs(title = "H13: Predicted vs observed") + theme_minimal()

# Participant-level MSE
p3 <- df_h13 |>
  group_by(id) |>
  summarise(across(all_of(str_replace(pred_cols, "predicted_d_", "err_")),
                   function(x) mean(x^2, na.rm = TRUE))) |>
  pivot_longer(-id, names_to = "model", values_to = "mse") |>
  mutate(model = str_remove(model, "^err_")) |>
  ggplot(aes(x = reorder(model, mse, median), y = mse)) +
  geom_boxplot(outlier.alpha = 0.15) +
  coord_flip() + ylim(0, 0.1) +
  labs(title = "H13: Participant-level MSE", x = NULL, y = "MSE") +
  theme_minimal()

p1; p2; p3
```

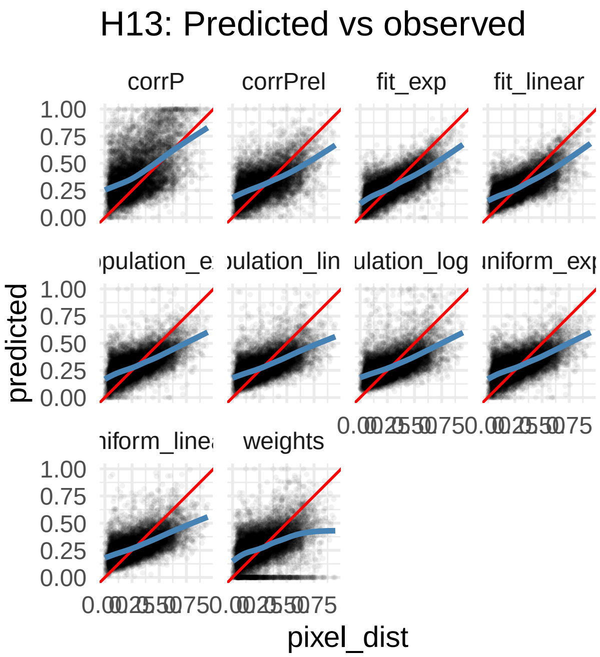
```
Warning: Removed 376 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

```
Warning: Removed 376 rows containing non-finite outside the scale range
(`stat_summary()`).
```

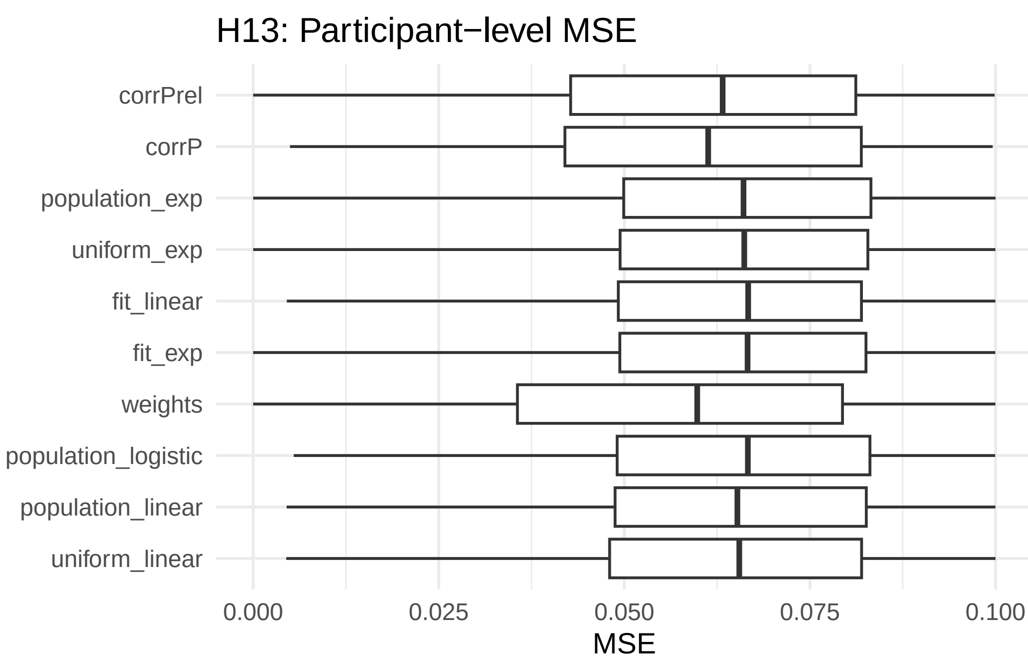



```
Warning: Removed 2335 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2335 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
Warning: Removed 5597 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



3.3.6.5 Explorative Analyses.

In addition to the preregistered hypotheses above, we explore several ways to define the function g_n (equation **??**) which describes the (participant-level) aggregation of belief differences on the six items q between individuals i and j into an overall belief distance.

In particular, we model different functional forms of g_n based on theories about who consider as social context and how this social context shapes their perception of the belief space—which we refer to as a social *lens*.

%\begin{mybox}{Exploration of different lens models} We define three different functional types of aggregation functions g_n with increasing level of complexity. First, a **simple lens** $\mathcal{S}_n$ is a 6-dimensional vector that scales (or weighs) each entry in the belief distance vector (equivalent to the weighted metric g_n in Box 1). Second, a **compression lens** $\mathcal{C}_n$ is a 6-dimensional vector, which projects the belief difference vector onto a single dimension, e.g.~`liberal' vs.`~conservative', and scales the belief distance in the direction of this dimension. Third, a **full lens** $\mathcal{F}_n$ is a 6×6 matrix that transforms the belief distance vector into a subjective new space with the same number of dimensions but with rotated and scaled axes. Writing these lenses in terms of transformation matrices yields:

1	simplified lens	$g_n(\lvert x_{n,i,q}-x_{n,j,q}\rvert) = \left\lVert \begin{pmatrix} & \mathcal{S}_n & 0 \\ 0 & & \end{pmatrix} \circ \begin{pmatrix} \lvert x_{n,i,1}-x_{n,j,1}\rvert \\ \vdots \\ \lvert x_{n,i,6}-x_{n,j,6}\rvert \end{pmatrix} \right\rVert_p$
2	compression lens	$g_n(\lvert x_{n,i,q}-x_{n,j,q\rvert}) = \left\lvert \begin{pmatrix} — & \mathcal{C}_n & — \\ 0 & \cdots & 0 \\ 0 & \vdots & 0 \end{pmatrix} \circ \begin{pmatrix} \lvert x_{n,i,1}-x_{n,j,1}\rvert \\ \vdots \\ \lvert x_{n,i,6}-x_{n,j,6}\rvert \end{pmatrix} \right\rvert$
3	full lens	$g_n(\lvert x_{n,i,q}-x_{n,j,q}\rvert) = \left\lVert \begin{pmatrix} \mathcal{F}_{n,11} & \cdots & \mathcal{F}_{n,61} \\ \vdots & \cdots & \vdots \\ \mathcal{F}_{n,61} & \cdots & \mathcal{F}_{n,66} \end{pmatrix} \circ \begin{pmatrix} \lvert x_{n,i,1}-x_{n,j,1}\rvert \\ \vdots \\ \lvert x_{n,i,6}-x_{n,j,6}\rvert \end{pmatrix} \right\rVert_p$

where $\|\cdot\|_p$ is the p-norm. Here, we choose Minkowski norm $p = 1$ or Euclidean norm $p = 2$.

The lenses $\mathcal{S}_n$, $\mathcal{C}_n$, and $\mathcal{F}_n$ of participant n depend on the belief distribution in the participant's social contexts, $x_{n,\text{context},q}$. We use two ways to define this social contexts:

- the beliefs of the seven to ten social circle contacts of the participant: $x_{n,\text{context},q} = \{x_{n,k,q} \mid k = 1 : K_n\}$.
- the beliefs of all the other participants in the participant's in-group, where in-group can mean (i) those who affiliate with the same party, $\{x_{m,\text{self},q} \mid m \text{ with party}_m = \text{party}_n\}$, (ii) those who place themselves in the same region on the left–right scale, (iii) those who have similar demographic attributes, or (iv) combinations thereof.

Then, we determine participant n's lenses …

- $\mathcal{S}_n$ as the inverse of the standard deviation of the $x_{n,\text{context},q}$. $\mathcal{S}_n = \left(\frac{1}{\text{Var}(x_{n,\text{context},1})}, \dots, \frac{1}{\text{Var}(x_{n,\text{context},6})}\right)$.
- $\mathcal{C}_n$ as the first (loaded) principal components of the social context beliefs. $\mathcal{C}_n = \frac{1}{\sqrt{\lambda_{n,1}}}\nu_{n,1}$.
- $\mathcal{F}_n$ as a principal component transformation, i.e as the inverse of the loaded principal component vectors of the social context beliefs. $\mathcal{F}_n = \left(\sqrt{\lambda_{n,1}}\,\nu_{n,1}, \dots, \sqrt{\lambda_{n,6}}\,\nu_{n,6}\right)^{-1}$, where $\nu_{n,\tilde{q}}$ are principal components and $\lambda_{n,\tilde{q}}$ the corresponding eigenvalues.

We predict map distances $d_{n,ij}$ using equation **??** with different choices of lenses and different definitions of social contexts. The goal of this exploration is to investigate to what extent these (theoretical) models and assumptions about participant-level social lenses describe how people perceive political differences between individuals.

```
print("later")
```

```
[1] "later"
```

3.4 Aim 4

Aim 4: Test the effects of an intervention on social circle belief variation

In wave 2, participants are randomly put in (i) a control condition, where they receive the same instruction as in wave 1, namely to choose seven to ten social contacts that they would count to their social circle, or (ii) a treatment condition, where they receive an additional instruction to choose contacts such that they represent a diverse mix of beliefs, if possible. Aim 4 is to test the effects of this instruction.

3.4.1 H14

3.4.1.1 H14. Participants in the treatment condition report higher social circle variance.

For the participants in the treatment condition T, who receive the instruction to choose diverse contacts, we measure the increase in social circle variance on issue q as

$$\Delta\text{var}_T = \sum_q \left(\mathbb{E}[\text{var}_{n,q}(w2) \mid \text{condition}_n = T] - \mathbb{E}[\text{var}_{n,q}(w1) \mid \text{condition}_n = T]\right) \tag{23}$$

The prediction is~$\Delta\text{var}_T > 0$ (one-sample t-test with significance threshold $\alpha = 0.05$) and $\Delta\text{var}_T > \Delta\text{var}_{\text{Control}}$ (using independent samples t-test with $\alpha = 0.05$). Social circle variance in the treatment condition T is higher than in wave 1 and higher than in the control group.

```
variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")

df_h14 <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(
    cols          = matches(paste0(var_col, "_")),
    names_to      = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, ")_(.*)")
  ) |>
```

```r
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "var_sc_wave", values_from = all_of(var_col)) |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control"),
         delta = var_sc_wave2 - var_sc_wave1)

cat("=== H14: Treatment increases social circle variance ===\n\n")

# Sanity check
df_h14 |>
  group_by(group) |>
  summarise(mean_w1 = mean(var_sc_wave1, na.rm = TRUE),
            sd_w1   = sd(var_sc_wave1,   na.rm = TRUE),
            mean_w2 = mean(var_sc_wave2, na.rm = TRUE),
            sd_w2   = sd(var_sc_wave2,   na.rm = TRUE),
            mean_d  = mean(delta,        na.rm = TRUE),
            n       = n()) |> print()

t_bal <- t.test(var_sc_wave1 ~ group, data = df_h14)
cat(sprintf("  Balance check (wave1 by group): t=%.3f  p=%.4f\n\n", t_bal$statistic, t_bal$p.value))

trt  <- df_h14 |> filter(group == "treatment")
ctrl <- df_h14 |> filter(group == "control")

# H14-1: Deltavar_T > 0
t_h14_1 <- t.test(trt$delta, mu = 0, alternative = "greater")
cat(sprintf("H14-1 (Deltavar_treatment > 0):\n"))
cat(sprintf("  Mean delta = %.4f ± %.4f\n", mean(trt$delta, na.rm = TRUE), sd(trt$delta, na.rm = TRUE)))
cat(sprintf("  t = %.4f,  p (one-sided) = %.4f\n", t_h14_1$statistic, t_h14_1$p.value))
cat(sprintf("  Supported : %s\n\n", ifelse(t_h14_1$statistic > 0 && t_h14_1$p.value < 0.05, "YES", "NO")))

# H14-2: Deltavar_T > Deltavar_Control
t_h14_2 <- t.test(delta ~ group, data = df_h14, alternative = "less", var.equal = FALSE)
cat(sprintf("H14-2 (Deltavar_treatment > Deltavar_control):\n"))
cat(sprintf("  Mean delta treatment = %.4f ± %.4f\n", mean(trt$delta,  na.rm = TRUE), sd(trt$delta,  na.rm = TRU
cat(sprintf("  Mean delta control   = %.4f ± %.4f\n", mean(ctrl$delta, na.rm = TRUE), sd(ctrl$delta, na.rm = TRU
cat(sprintf("  Difference (T - C)   = %.4f\n", mean(trt$delta, na.rm = TRUE) - mean(ctrl$delta, na.rm = TRUE)))
cat(sprintf("  t = %.4f,  p (one-sided) = %.4f\n", t_h14_2$statistic, t_h14_2$p.value))
cat(sprintf("  Supported : %s\n\n", ifelse(t_h14_2$statistic < 0 && t_h14_2$p.value < 0.05, "YES", "NO")))
```

```
=== H14: Treatment increases social circle variance ===

# A tibble: 2 x 7
  group     mean_w1 sd_w1 mean_w2 sd_w2  mean_d     n
  <chr>       <dbl> <dbl>   <dbl> <dbl>   <dbl> <int>
1 control     0.268 0.198   0.275 0.197 0.00638  2370
2 treatment   0.285 0.199   0.311 0.205 0.0262   3042
  Balance check (wave1 by group): t=-3.044  p=0.0023

H14-1 (Deltavar_treatment > 0):
  Mean delta = 0.0262 ± 0.1884
  t = 7.6679,  p (one-sided) = 0.0000
  Supported : YES

H14-2 (Deltavar_treatment > Deltavar_control):
  Mean delta treatment = 0.0262 ± 0.1884
  Mean delta control   = 0.0064 ± 0.1780
  Difference (T - C)   = 0.0198
  t = -3.9599,  p (one-sided) = 0.0000
  Supported : YES
```

```r
df_h14_2 <- df_h14 |>
  left_join(df_p|>filter(wave==2)|>select(id,time_between_waves), by = c("id"))|>
  mutate(time_between_waves_z = scale(time_between_waves))
m <- lm(delta ~ group + time_between_waves_z, data = df_h14_2)
```

```
tidy_print(m)
summary(m)$r.squared
```

```
# A tibble: 3 x 4
  term                estimate std.error p.value
  <chr>                  <dbl>     <dbl>   <dbl>
1 (Intercept)            0.006     0.004   0.091
2 grouptreatment         0.02      0.005   0
3 time_between_waves_z   0.001     0.003   0.578
```

term	estimate	std.error	p.value
(Intercept)	0.006	0.004	0.091
grouptreatment	0.020	0.005	0.000
time_between_waves_z	0.001	0.003	0.578

```
[1] 0.002906804
```

3.4.2 H15

3.4.2.1 H15. Higher relative social circle variance decreases issue weight.

We model the new issue weights, $\alpha_{n,q}(w2)$, inferred in wave 2, of the participants in the treatment condition as

$$\alpha_{n,q}(w2) = \alpha_{n,q}(w1) + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \frac{\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)}{\sum_q \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)} + \epsilon_{n,q} \tag{24}$$

The prediction is $\beta_1 < 0$. Higher relative social circle variance on one issue decreases the weight of that issue for the participant. %We test this via a t-test with significance threshold $\alpha = 0.05$.

UPDATE: There should be abs signs in the denominator! Otherwise small delta Var's are going to blow up things dramatically.
UPDATE: This does not need to depend on the Treatment condition group alone. I analysed both now
OPEN QUESTION: Is the denominator: $\sum_q |\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)|$ or $\sum_q |\text{var}_{n,q}(w1)|$?? I chose the first now because this is what we preregistered.

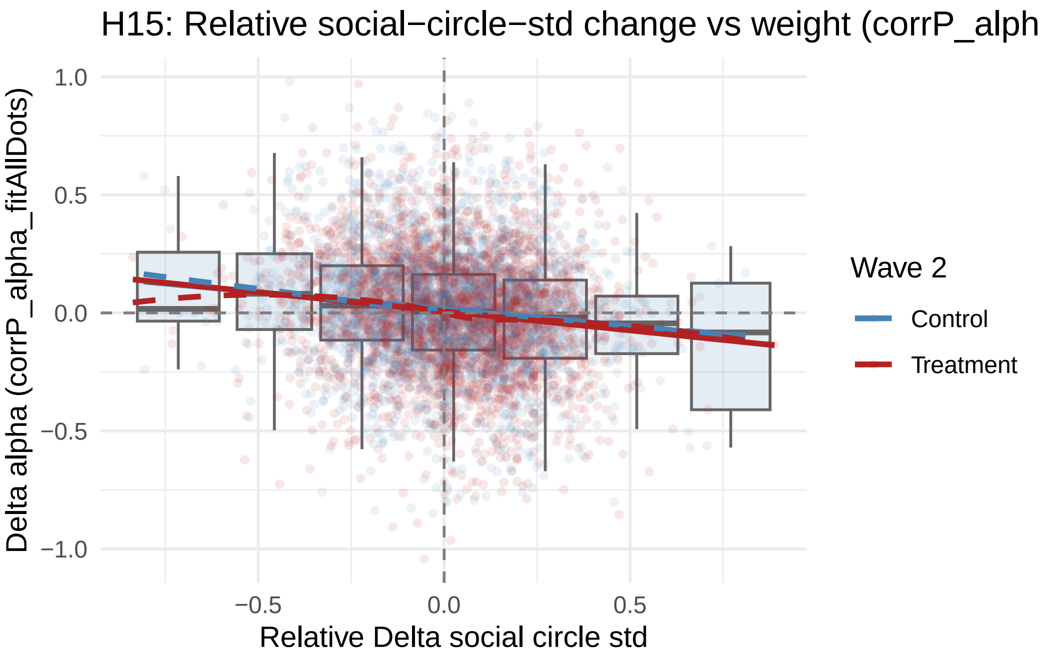
```
variance_type <- "std"
k             <- paste0("corrP_alpha_", fitmode)
var_col       <- paste0(variance_type, "_socialCircle_ops")

# Variance change per participant per issue
df_h15_var <- df_p |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
               names_to  = c(".value", "issue"),
               names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  rename(var_sc = all_of(var_col)) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "w", values_from = var_sc) |>
  rename(var_w1 = w1, var_w2 = w2) |>
  mutate(delta_var = var_w2 - var_w1) |>
  group_by(id) |>
  mutate(
    sum_delta_var = sum(abs(delta_var), na.rm = TRUE),
    #  denominator is Σ|Deltavar| (preregistered); see note in LaTeX about alternatives
    rel_delta_var = if_else(sum_delta_var > 0.05, delta_var / sum_delta_var, NA_real_)
  ) |>
  ungroup()

# Alpha weights across waves
df_h15_alpha <- df_p |>
  pivot_longer(cols = matches(paste0("^", k)),
               names_to  = c(".value", "issue"),
               names_pattern = paste0("^(", k, ")_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "alpha_w", values_from = all_of(k))
```

```r
df_h15 <- df_h15_var |>
  left_join(df_h15_alpha, by = c("id", "issue")) |>
  left_join(df_p |> filter(wave == 2) |> select(id, treatment_wave2), by = "id")

m_h15 <- lm(alpha_w2 ~ alpha_w1 + rel_delta_var , data = df_h15)
cf_h15 <- summary(m_h15)$coefficients["rel_delta_var", ]

print_one_sided(
  sprintf("H15: Relative Delta social circle %s decreases issue weight (%s)", variance_type, k),
  cf_h15["Estimate"], cf_h15["Std. Error"], cf_h15["t value"],
  df   = m_h15$df.residual,
  direction = "negative"
)
```

```
=== H15: Relative Delta social circle std decreases issue weight (corrP_alpha_fitAllDots) ===
  Estimate  =  -0.1078
  SE        =   0.0146
  t         =  -7.3689
  df        = 5363.0
  p (one-sided, beta < 0) = 0.0000
  Supported : YES
```

```r
df_h15 |>
  filter(!is.na(rel_delta_var), !is.na(alpha_w2), !is.na(alpha_w1)) |>
  mutate(
    delta_alpha       = alpha_w2 - alpha_w1,
    rel_delta_var_bin = cut(rel_delta_var, breaks = 7),
    treatment_wave2   = factor(treatment_wave2, levels = c(0, 1),
                               labels = c("Control", "Treatment"))
  ) |>
  ggplot(aes(x = rel_delta_var, y = delta_alpha, color = treatment_wave2)) +
  geom_boxplot(aes(group = rel_delta_var_bin), fill = "steelblue", alpha = 0.15,
               outlier.shape = NA, color = "gray40") +
  geom_point(alpha = 0.1, size = 1) +
  geom_smooth(formula = y ~ x, method = "lm",    se = FALSE, alpha = 0.5) +
  geom_smooth(formula = y ~ x, method = "loess", se = FALSE, linetype = "dashed", alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  scale_color_manual(values = c("Control" = "steelblue", "Treatment" = "firebrick"),
                     name = "Wave 2") +
  labs(title = sprintf("H15: Relative social-circle-%s change vs weight (%s) change",
                       variance_type, k),
       x = sprintf("Relative Delta social circle %s", variance_type),
       y = sprintf("Delta alpha (%s)", k)) +
  theme_minimal()
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```



3.4.3 H16 (caution)

3.4.3.1 H16. Increase in overall social circle variance decreases distance to typical voters.

We model the map distances between the participant themselves and typical voters of political parties r in wave $w2$ as

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \mathbb{1}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n,\text{self}-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + \epsilon_{n,r} \tag{25}$$

The explanatory variable is the change in social circle variance averaged over all six issues and weighted by the corresponding belief differences between participant and the typical voter of party r, $\Delta x_{n,\text{self}-r,q} = |x_{n,\text{self},q} - x_{n,r,q}|$.

The prediction is $\beta_1 < 0$. When overall social circle variance increases, the reported distance between participant and typical voter beliefs decreases. %We test this via a t-test with significance threshold $\alpha = 0.05$.

ERROR: This should not include the treatment. The effect we want to test is whether variance increase (weighted by opinion difference, such that) changes distances.

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \sum_q \Delta x_{n,\text{self}-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + \beta_2 \cdot condition_n + u_n + \epsilon_{n,r} \tag{26}$$

or alternatively, but that's not really a fair thing to test and not really related to our theory (see discussion with Mirta):

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot condition_n + \epsilon_{n,r} \tag{27}$$

```
# Build shared data objects used by both H16 and H17 variants

variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")

# Wave 1 and 2 self-party distances
df_self_party_w1 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 1) |>
  select(id, dot2, pixel_dist_w1 = pixel_dist)

df_self_party_w2 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, pixel_dist_w2 = pixel_dist, sympathy_w2 = sympathy)

# deltaX (wave 2)
df_deltaX <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, all_of(col_deltas))

# Variance change per participant per issue (wide)
```

```r
df_delta_var_wide <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
               names_to  = c(".value", "issue"),
               names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "var_w", values_from = all_of(var_col)) |>
  mutate(delta_var = var_w2 - var_w1) |>
  pivot_wider(id_cols = id, names_from = issue,
              names_prefix = "delta_var_", values_from = delta_var)

# Long delta_var (for new H16/H17 moderation models)
df_delta_var_long <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
               names_to  = c(".value", "issue"),
               names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "var_w", values_from = all_of(var_col)) |>
  mutate(delta_var   = var_w2 - var_w1,
         delta_var_c = as.numeric(scale(delta_var, center = TRUE, scale = FALSE))) |>
  select(id, issue, delta_var, delta_var_c)
```

```r
# Simple direct-effect check (does treatment affect pixel_dist?)
df_h16_simple <- df_self_party_w2 |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control")) |>
  left_join(df_self_party_w1, by = c("id", "dot2"))

m_h16_direct <- lmer(pixel_dist_w2 ~ group + (1 | id) + (1 | dot2),
                     data = df_h16_simple, REML = TRUE)
cat("=== H16 (simple: treatment → pixel_dist) ===\n")
print(summary(m_h16_direct)$coefficients)

# Full model: weighted variance change
df_h16_full <- df_deltaX |>
  left_join(df_self_party_w1,   by = c("id", "dot2")) |>
  left_join(df_self_party_w2,   by = c("id", "dot2")) |>
  left_join(df_delta_var_wide,  by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(col_deltas)) * across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    )
  )

m_h16_full <- lmer(pixel_dist_w2 ~ pixel_dist_w1 + weighted_var_change + group + (1 | id),
                   data = df_h16_full, REML = TRUE)
cf_h16 <- lmer_coef(m_h16_full, "weighted_var_change")

cat("\n")
print_one_sided("H16 (full): weighted Deltavar decreases perceived distance",
                cf_h16$beta, cf_h16$se, cf_h16$t, direction = "negative")
```

```
=== H16 (simple: treatment → pixel_dist) ===
                   Estimate  Std. Error         df    t value     Pr(>|t|)
(Intercept)     0.338015954 0.017448847   7.643035 19.3718221 8.988148e-08
grouptreatment -0.006245272 0.008341309 900.000072 -0.7487161 4.542240e-01

=== H16 (full): weighted Deltavar decreases perceived distance ===
  Estimate  =   0.0082
  SE        =   0.0046
  t         =   1.7734
  df        = infty
```

```
  p (one-sided, beta < 0) = 0.9619
  Supported : NO
```

3.4.4 H17 (caution)

3.4.4.1 H17. Increase in overall social circle variance increases likeability rating of typical voters.

Similarly, we model how participant n rates the likeability rating of typical voters of party r in wave 2 as

$$y_{n,r} = \beta_0 + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n,\text{self}-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + u_n + \epsilon_{n,r} \tag{28}$$

The prediction is $\beta_1 > 0$. When overall social circle variance increases, the likeability rating of typical voters of the political parties increases. %We test this via a t-test with significance threshold $\alpha = 0.05$.

Most simple: Does treatment affect liking? No obviously not!

```
# Simple direct-effect check
df_h17_simple <- df_self_party_w2 |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control"))

m_h17_direct <- lmer(sympathy_w2 ~ group + (1 | id) + (1 | dot2),
                     data = df_h17_simple, REML = TRUE)
cat("=== H17 (simple: treatment → sympathy) ===\n")
print(summary(m_h17_direct)$coefficients)

# Full model
df_h17_full <- df_deltaX |>
  left_join(df_self_party_w2,  by = c("id", "dot2")) |>
  left_join(df_delta_var_wide, by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(col_deltas)) * across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    )
  )

m_h17_full <- lmer(sympathy_w2 ~ weighted_var_change + group + (1 | id),
                   data = df_h17_full, REML = TRUE)
cf_h17 <- lmer_coef(m_h17_full, "weighted_var_change")

cat("\n")
print_one_sided("H17 (full): weighted Deltavar increases likeability",
                cf_h17$beta, cf_h17$se, cf_h17$t, direction = "positive")
```

```
=== H17 (simple: treatment → sympathy) ===
                   Estimate  Std. Error         df     t value     Pr(>|t|)
(Intercept)     0.423891501 0.026777532   6.507399 15.8301182 1.939544e-06
grouptreatment -0.003302603 0.008775184 899.999845 -0.3763571 7.067402e-01

=== H17 (full): weighted Deltavar increases likeability ===
  Estimate  =  -0.0302
  SE        =   0.0075
  t         =  -4.0329
  df        = infty
  p (one-sided, beta > 0) = 1.0000
  Supported : NO
```

3.5 NEW: new versions of H16 and H17 (moderation)

3.5.1 H16 new

3.5.1.1 H16: Treatment increases social circle variance, which weakens the effect of opinion differences on perceived distance.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q} = |x_{n,\text{self},q} - x_{n,r,q}|$ and perceived distance d? We model:

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{29}$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 < 0$: the treatment weakens the effect of opinion differences on perceived distance.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta\text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \to d$ relationship, above and beyond treatment assignment?

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta\text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta\text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{30}$$

The prediction is $\beta_3 < 0$: when social circle variance increases, the effect of opinion differences on perceived distance weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta\text{var} \xrightarrow{\beta_3<0} \text{weakens } (\Delta X \to d) \tag{31}$$

```r
# Step 1 - ITT: does treatment moderate DeltaX → distance?
df_h16_itt <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2,
         id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0("^(deltaX|", var_col, ")_(", paste(questions_sc, collapse = "|"), ")$")),
               names_to  = c(".value", "issue"),
               names_pattern = paste0("^(deltaX|", var_col, ")_(.*)")) |>
  select(id, wave, dot2, pixel_dist, issue, deltaX, all_of(var_col)) |>
  mutate(treatment = as.integer(id %in% treatment_ids))

m_h16_itt <- lmer(pixel_dist ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 | dot2),
                  data = df_h16_itt)
cf_itt_16  <- lmer_coef(m_h16_itt, "deltaX:treatment")

cat("\n")
print_one_sided("H16-new Step 1 (ITT): treatment weakens DeltaX → distance",
                cf_itt_16$beta, cf_itt_16$se, cf_itt_16$t, direction = "negative")

# Step 2 - Deltavar as moderator
df_h16_var <- df_h16_itt |>
  left_join(df_delta_var_long, by = c("id", "issue"))

m_h16_var  <- lmer(pixel_dist ~ deltaX * delta_var_c + (1 | id) + (1 | issue) + (1 | dot2),
                   data = df_h16_var)
cf_var_16  <- lmer_coef(m_h16_var, "deltaX:delta_var_c")

print_one_sided("H16-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → distance",
                cf_var_16$beta, cf_var_16$se, cf_var_16$t, direction = "negative")
```

```
=== H16-new Step 1 (ITT): treatment weakens DeltaX → distance ===
  Estimate  =   0.0044
  SE        =   0.0030
  t         =   1.4548
  df        = infty
  p (one-sided, beta < 0) = 0.9271
  Supported : NO

=== H16-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → distance ===
  Estimate  =   0.0035
  SE        =   0.0071
  t         =   0.4890
  df        = infty
  p (one-sided, beta < 0) = 0.6876
  Supported : NO
```

3.5.2 H17 new

3.5.2.1 H17: Treatment increases social circle variance, which increases likeability ratings of typical party voters.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q}$ and likeability ratings $y_{n,r}$? We model:

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (32)$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 > 0$: the treatment weakens the negative effect of opinion differences on likeability, i.e. participants in the treatment condition rate typical voters of parties they disagree with more favourably.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta\text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \to y$ relationship, above and beyond treatment assignment?

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta\text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta\text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (33)$$

The prediction is $\beta_3 > 0$: when social circle variance increases, the negative effect of opinion differences on likeability weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta\text{var} \xrightarrow{\beta_3 > 0} \text{weakens } (\Delta X \to -y) \quad (34)$$

```
df_h17_itt <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2,   # fixed: was "2"
         id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0("^(deltaX|", var_col, ")_(", paste(questions_sc, collapse = "|"), ")$")),
               names_to  = c(".value", "issue"),
               names_pattern = paste0("^(deltaX|", var_col, ")_(.*)")) |>
  select(id, wave, dot2, sympathy, issue, deltaX, all_of(var_col)) |>
  mutate(treatment = as.integer(id %in% treatment_ids))

m_h17_itt <- lmer(sympathy ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 | dot2),
                  data = df_h17_itt)
cf_itt_17  <- lmer_coef(m_h17_itt, "deltaX:treatment")

cat("\n")
print_one_sided("H17-new Step 1 (ITT): treatment weakens DeltaX → low likeability",
                cf_itt_17$beta, cf_itt_17$se, cf_itt_17$t, direction = "positive")

df_h17_var <- df_h17_itt |>
  left_join(df_delta_var_long, by = c("id", "issue"))

m_h17_var <- lmer(sympathy ~ deltaX * delta_var_c + (1 | id) + (1 | issue) + (1 | dot2),
                  data = df_h17_var)
cf_var_17 <- lmer_coef(m_h17_var, "deltaX:delta_var_c")

print_one_sided("H17-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → low likeability",
                cf_var_17$beta, cf_var_17$se, cf_var_17$t, direction = "positive")
```

```
=== H17-new Step 1 (ITT): treatment weakens DeltaX → low likeability ===
  Estimate  =  -0.0168
  SE        =   0.0053
  t         =  -3.1633
  df        = infty
  p (one-sided, beta > 0) = 0.9992
  Supported : NO

=== H17-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → low likeability ===
  Estimate  =  -0.0256
```

```
SE        =   0.0126
t         =  -2.0382
df        = infty
p (one-sided, beta > 0) = 0.9792
Supported : NO
```

#

4 Exclusion criteria

Participants are excluded if they complete the survey in less than one third of the median completion time, provide invariant slider responses, or do not indicate their own beliefs on the six political issues.